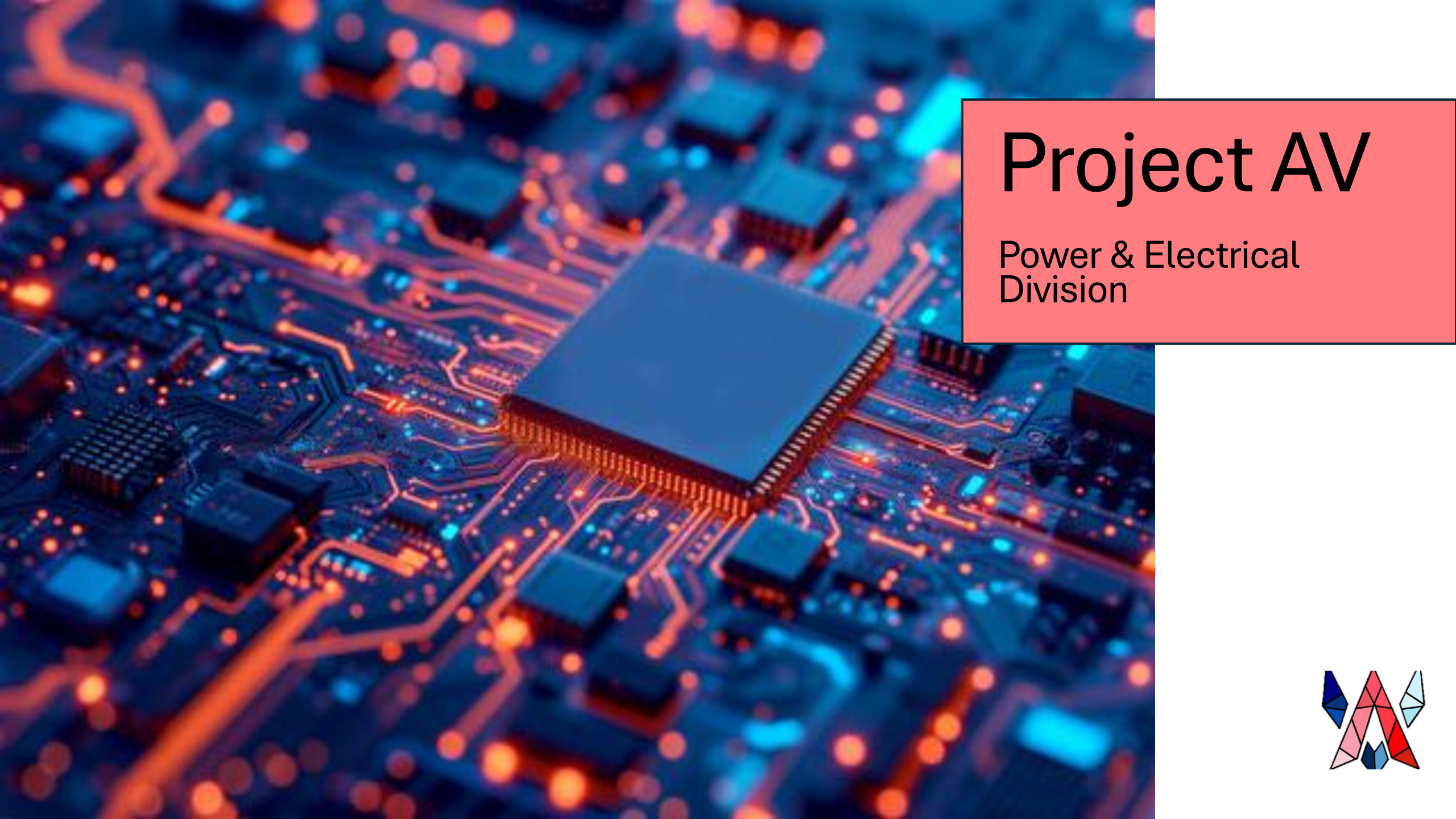




Project AV

Power & Electrical
Division

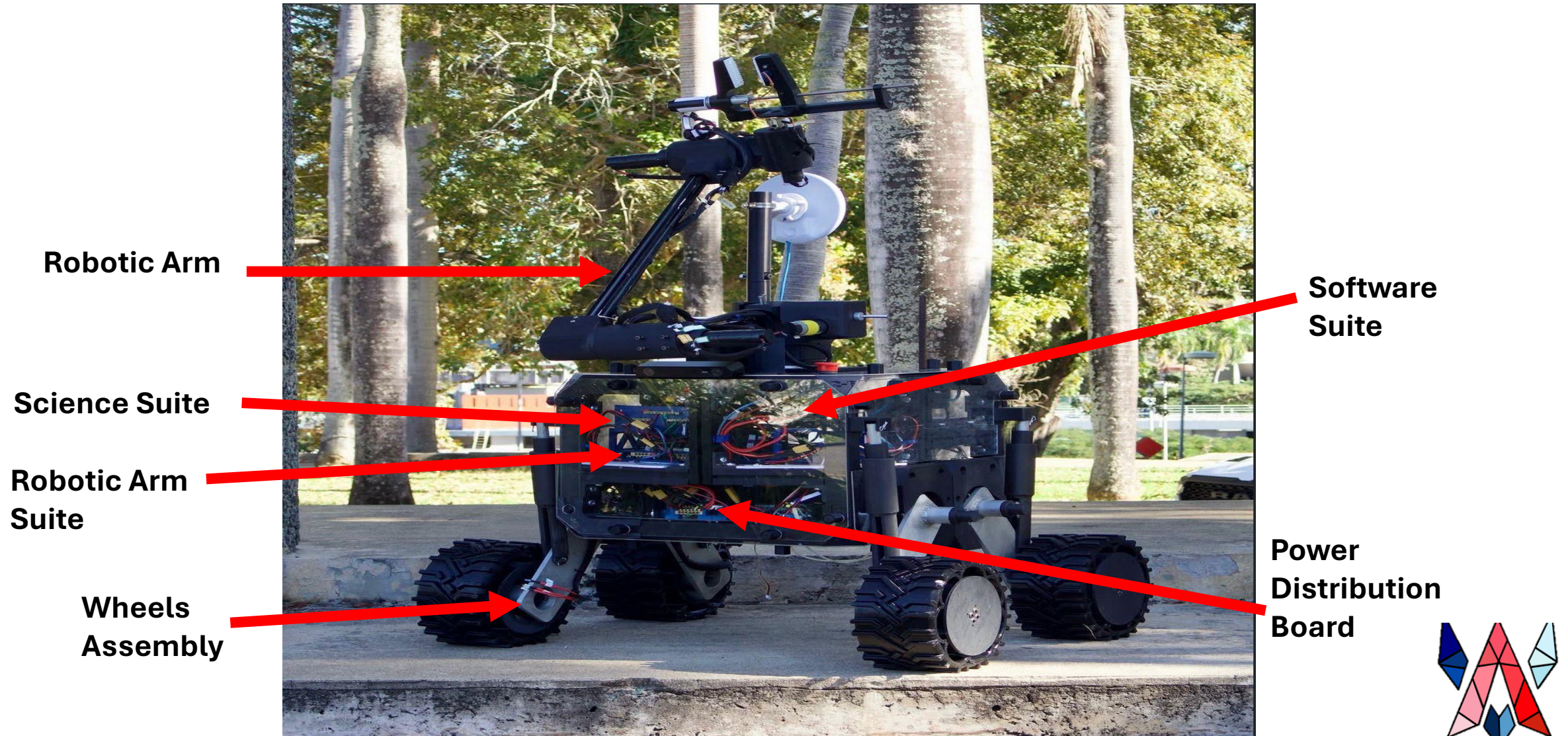


Areas of Improvement for B0R1

- Telemetry, BMS & Power Distribution Board – Sebastián Pagán
- Battery Testing, Performance & Wheels – Christian Arroyo
- Robotic Arm Suite – Christopher Morales & Anyaris Quiñones
- Wiring Layout & Design – Andrés Irizarry
- Science Suite – Christian Rexach



Rover Under Study



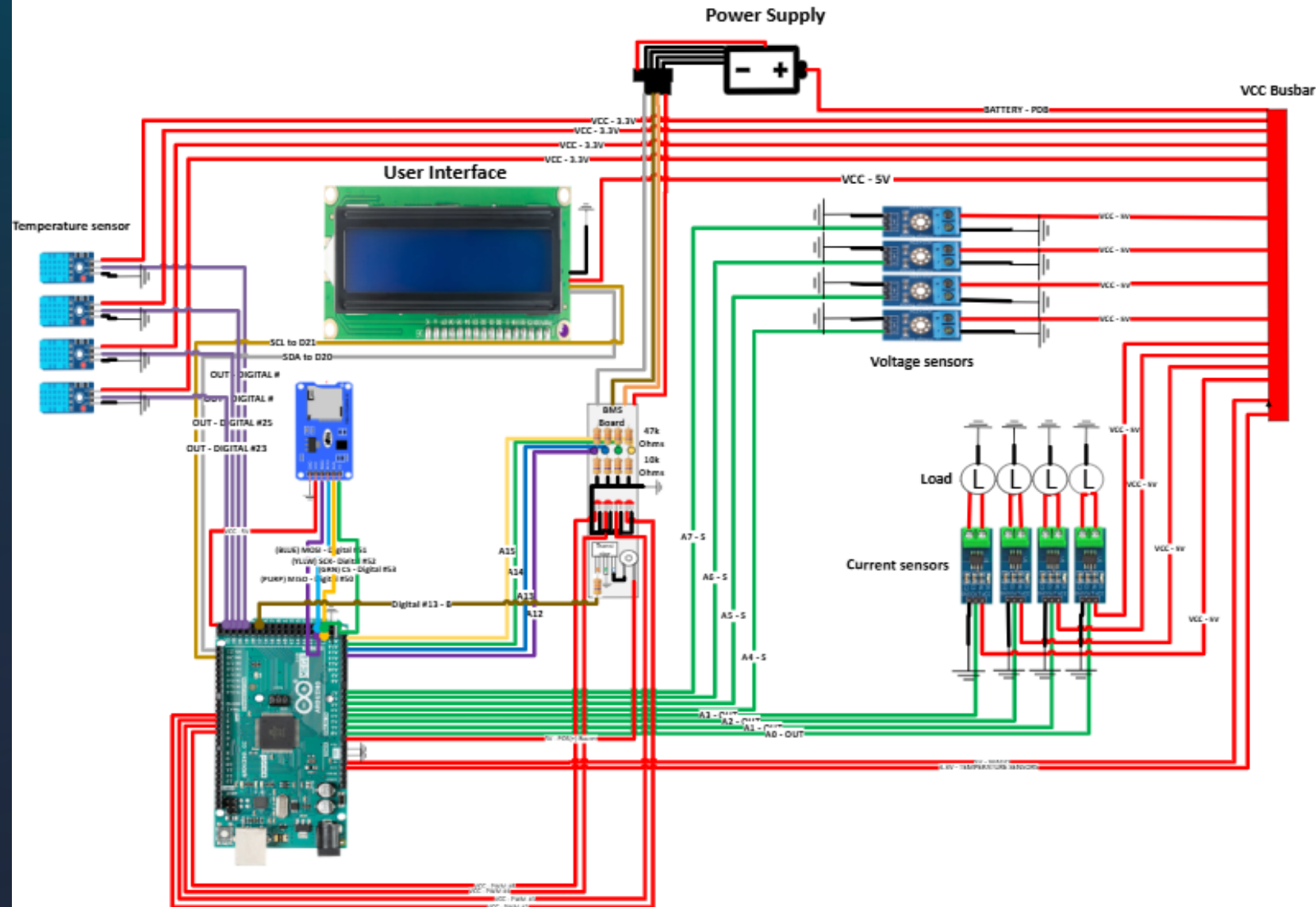
+
◦ • Telemetry, BMS &
Distribution Board



Telemetry System Schematic as of 5/4/26

NOTES

Busbars are used to represent common nodes between jumper cables

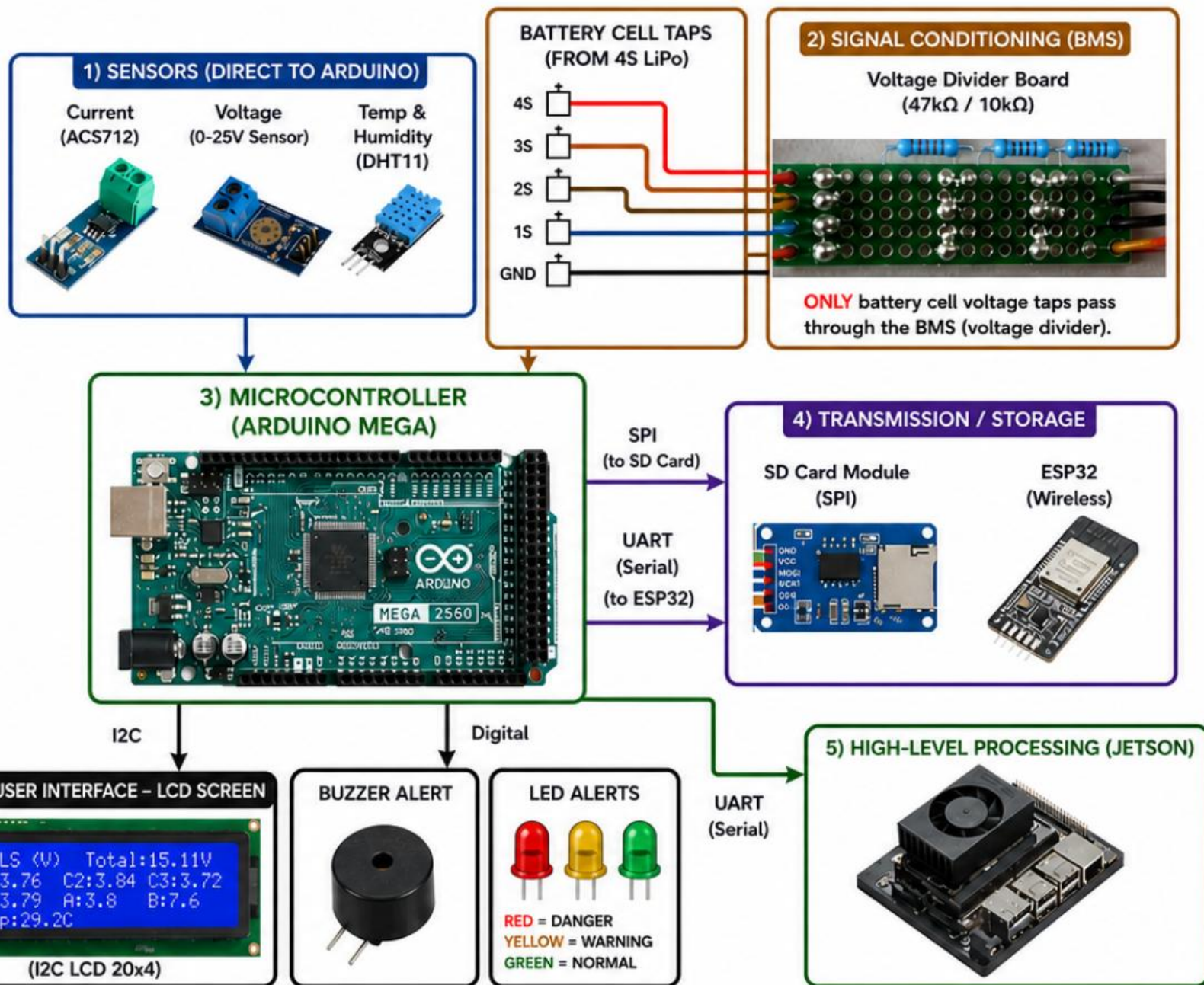


What is Telemetry?

Telemetry is the process of acquiring, transmitting and analyzing data from remote systems for monitoring and decision-making.

Components of a Telemetry System

- 1. Sensors (Direct to Arduino)**
Measure physical parameters such as current, voltage, temperature and humidity.
- 2. Signal Conditioning (BMS)**
ONLY battery cell voltage taps go through the BMS (voltage divider) to scale high voltages to safe 0–5V levels for the Arduino.
- 3. Microcontroller (Arduino Mega)**
Reads all sensor data, processes it, controls outputs and manages data logging and transmission.
- 4. Transmission / Storage**
Sends data to external systems or logs locally.
- 5. High-Level Processing (Jetson)**
Performs advanced analysis, visualization and decision-making.
- 6. User Interface**
Displays data and alerts to the user (LCD, buzzer, dashboard).



Data Flow: Sensors (direct) & Battery Taps (via BMS) → Arduino (Processing) → Transmission / Storage → Jetson (Analysis) → User Interface

Telemetry System



Challenges

- Limited real-time fault awareness
- Local data can be harder to access quickly
- Operators need faster visual and audible alerts

Proposition

- Add SD / ESP32 logging for local and wireless monitoring
- Add buzzer and LED alerts for immediate fault indication
- Keep the system modular and easy to expand

Telemetry System

Active Buzzer vs Passive Buzzer

Active Buzzer



✓ Pros

- Simple control: just `digitalWrite(pin, HIGH) / LOW`
- More reliable for critical alarms
- Loud and consistent alert
- Easier and faster to implement

✗ Cons

- Usually only one tone
- Less flexible for coded fault patterns
- Fewer customization options

Passive Buzzer



✓ Pros

- Can generate different tones with `tone()`
- Good for coded alerts (overvoltage, overtemp, etc.)
- More flexible and customizable

✗ Cons

- Requires `tone()` and timing control
- Can behave inconsistently if the loop timing changes
- Sometimes quieter depending on the model
- Slightly more software complexity



Recommendation for Rover Telemetry

Use an active buzzer for the main critical alarm because it is simpler and more reliable.
Use a passive buzzer only if different sound patterns are needed for multiple fault types.

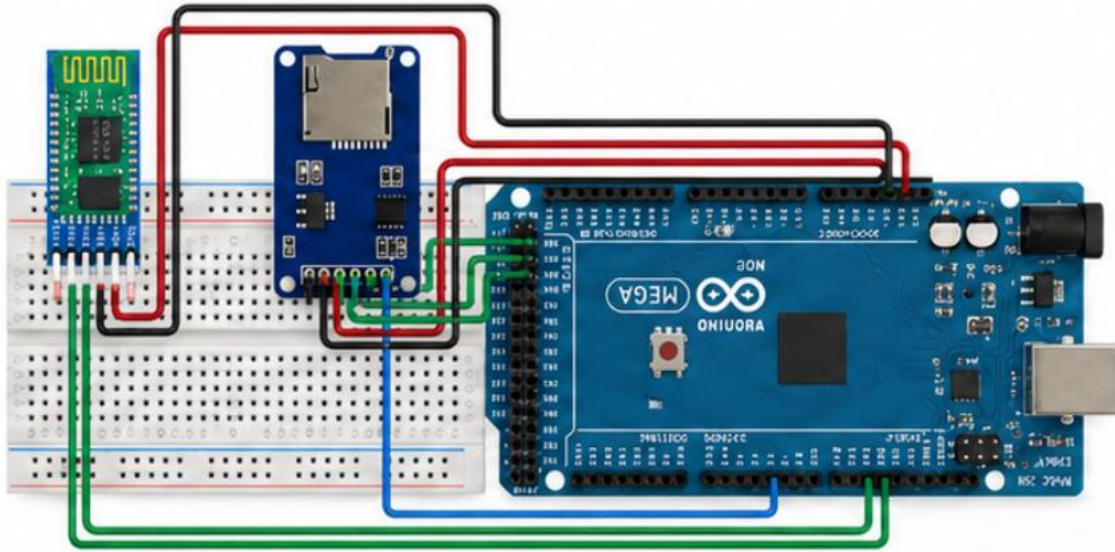


Examples: OV = fast beep | Overtemp = slow beep

Data Loggers

Wired SD Card Module vs Wireless Bee Data Logger

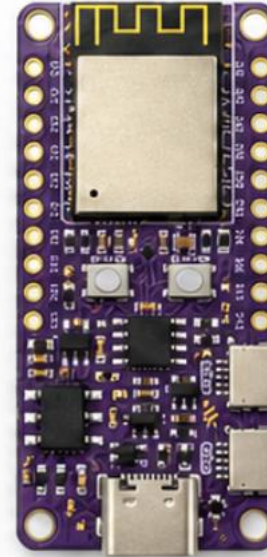
Wired SD Card Module



- Direct wired logging to Arduino Mega
- Simple and reliable local data storage
- Best for basic black-box logging

- ✓ **Pros:** simple, inexpensive, reliable
- ✗ **Cons:** more wiring, no built-in wireless telemetry

Wireless Bee Data Logger



Bee Data Logger



- Wireless-capable data logging platform
- Compact all-in-one logger for telemetry applications
- Best for remote monitoring and flexible expansion

- ✓ **Pros:** wireless, compact, expandable
- ✗ **Cons:** more setup complexity, higher cost

Data Logger Differences

Purpose	Wired SD Card Module	Wireless Bee Data Logger
Main role	Local black box logging	Wireless telemetry + logging
Reliability	Very reliable	Good, but wireless can fail
Complexity	Low	Medium/high
Wiring	More direct wiring to Arduino	Cleaner if used as all-in-one, but needs communication setup
Timestamps	Needs external RTC	Built-in RTC
Wireless	No	Yes, WiFi/Bluetooth capable
Best for URC	Backup fault log	Live monitoring/dashboard
Risk	SD corruption during power loss	Network/code/power complexity

BMS or Voltage Divider?

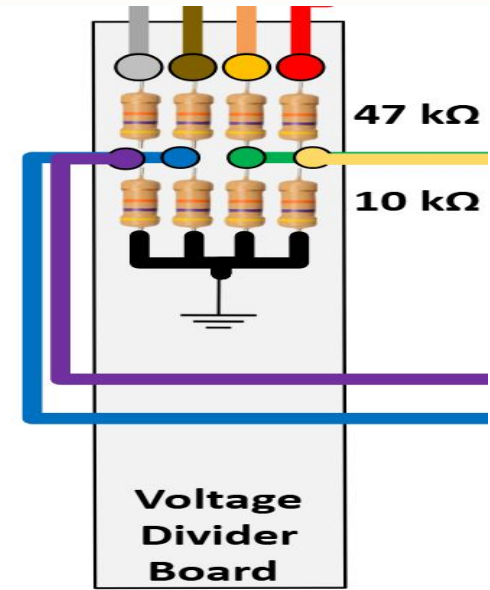
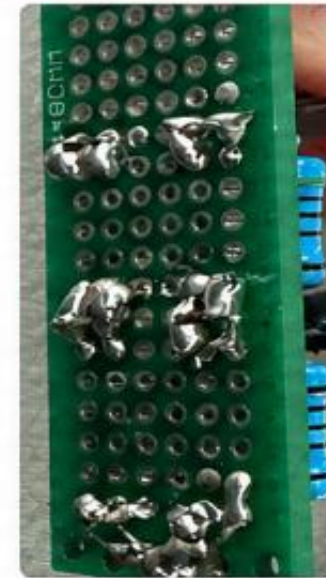
Our current board is a voltage divider used for 4S LiPo cell-voltage monitoring — not a full BMS.

What a Full BMS Typically Does

- ✓ Disconnects the battery during a fault
- ✓ Balances cells
- ✓ Stops charging when limits are reached
- ✓ Limits overcurrent
- ✓ Protects against short circuits

i *Telemetry logging is handled by the Arduino / SD system, not by the voltage divider board.*

What Our Current Board Does (May 2, 2026)



47 kΩ / 10 kΩ resistor divider network

- Reads 4S LiPo battery cell taps
- Scales high voltages to a safe range for the Arduino
- Supports cell-voltage monitoring only

★ Conclusion: This board is a voltage divider for monitoring, not a full battery management system (BMS).

Key Components and Recommended Upgrades

Improving safety, accuracy, and usability

1. Core Voltage Divider Components



**47kΩ Resistor
(1%)**

Sets the main divider ratio for each cell tap.



**10kΩ Resistor
(1%)**

Lower-leg resistor that completes the divider.



**5.1V Zener
or TVS Diode**

Clamps the tap voltage to ~5.1V to protect the ADC.



0.1μF Capacitor

Filters high-frequency noise for stable and smooth readings.



**JST-XH Balance
Connector (5-pin)**

Provides a safe and reliable connection to the battery.



Enclosure

Protects the board from shorts, dust, and physical damage.

2. Alert and Interface Additions



Green LED

Normal system condition indicator.



Yellow LED

Warning condition indicator.



Red LED

Danger or critical condition indicator.



**220Ω or 330Ω
Resistors**

Current limiting for LEDs and indicator circuits.



Active Buzzer

Audible alert for warnings or critical events.



**NPN Transistor or
Logic MOSFET**

Drives buzzer or other loads safely from the Arduino.



**Panel-Mount
Holders**

Secure and professional mounting on the enclosure.



**Best upgrades
for this board**

1

**1% resistors +
JST-XH connector**

2

**5.1V protection
diodes**

3

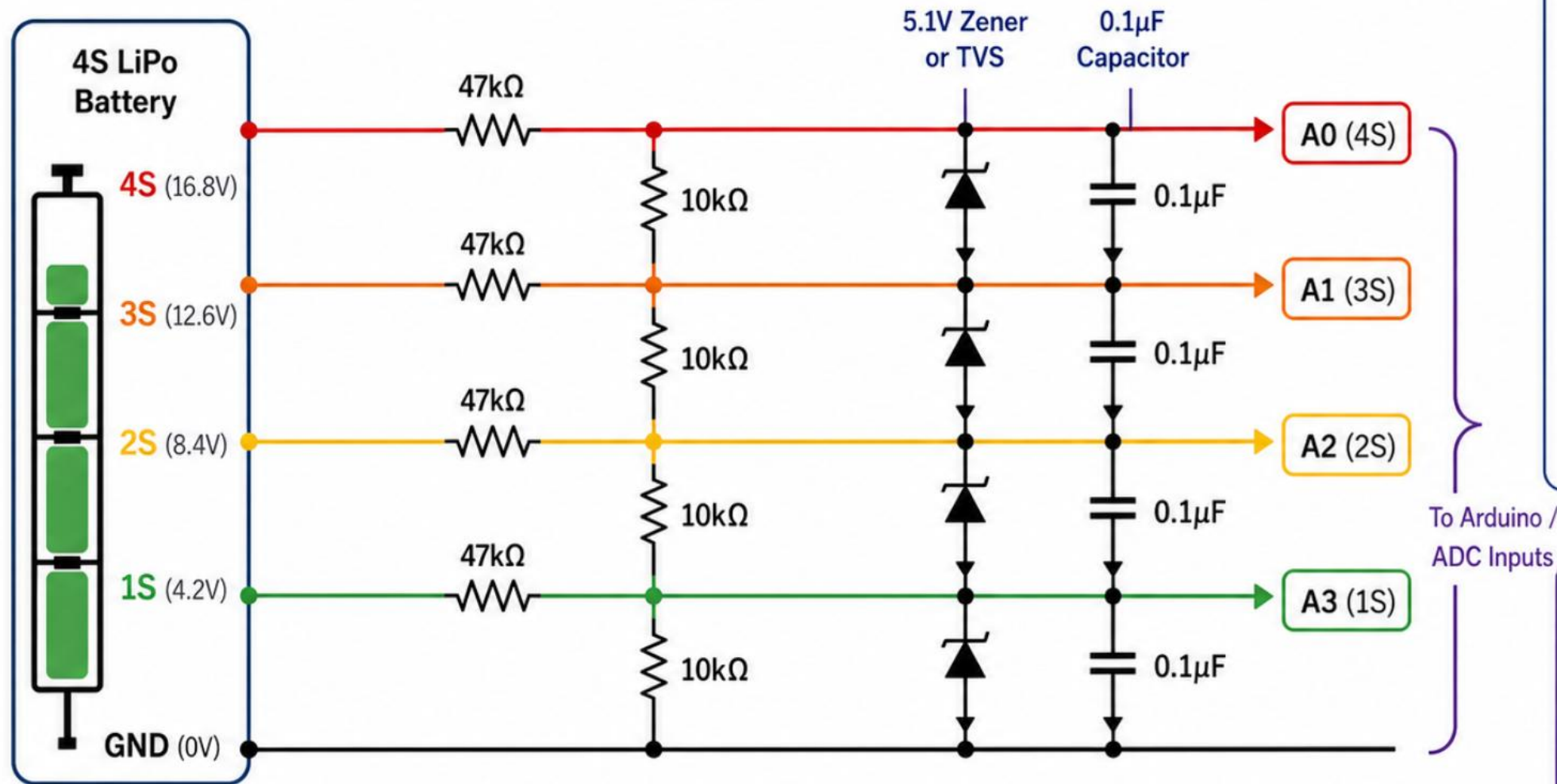
**LED alert system +
active buzzer**



This board remains a voltage divider for monitoring only, not a full BMS.

Voltage Divider Board Schematic

4S LiPo cell-voltage monitoring



How it Works

- **Resistor divider** scales each cell tap down for safe ADC measurement.
- **Zener/TVS** limits voltage during faults or spikes to protect the ADC.
- **Capacitor** filters noise for stable and accurate readings.

ADS1115 optional for improved measurement accuracy.

Board is for monitoring, not full BMS protection.

Zener Diode vs TVS Diode

Protection options for the voltage divider board

5.1V Zener Diode



Simple voltage clamp for low-power signal protection.

Pros

- ✓ Low cost
- ✓ Simple and widely available
- ✓ Good for basic analog-input protection
- ✓ Works well as a backup clamp on the divider output

Cons

- ✗ Slower than a TVS diode
- ✗ Less robust against large spikes
- ✗ Not ideal for harsh transient events

5.0V–5.6V TVS Diode



Fast transient suppression for spikes and ESD-like events.

Pros

- ✓ Faster response
- ✓ Better for electrical spikes and transients
- ✓ More rugged protection
- ✓ Good for noisy rover electronics

Cons

- ✗ Usually more expensive
- ✗ Can be physically larger
- ✗ May be more than needed for a simple low-energy line



Recommendation for this board: A 5.1V Zener diode is the simpler low-cost choice for basic Arduino input protection; a TVS diode is the better choice if stronger spike protection is desired.

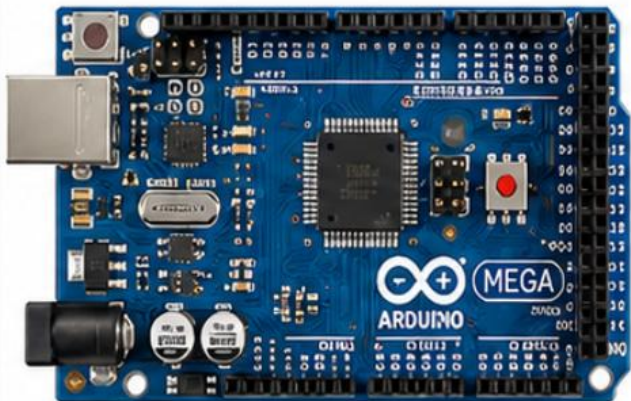


These parts protect the Arduino input line only; they do not replace the resistor divider.

ADS1115 vs Arduino Mega

Battery voltage measurement options for a 4S LiPo telemetry system

Arduino Mega (Built-in ADC)



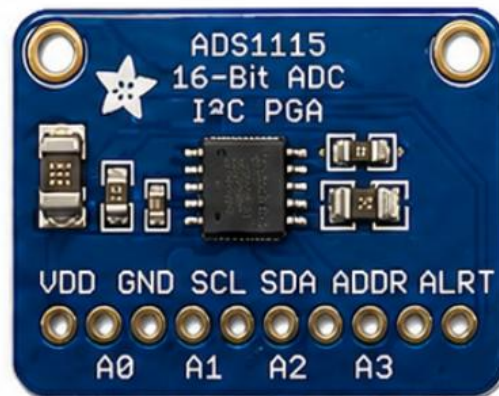
Pros:

- ✓ Already part of the system
- ✓ No extra hardware required
- ✓ Simple wiring
- ✓ Good for prototype testing

Cons:

- ✗ 10-bit ADC resolution
- ✗ More noise in analog readings
- ✗ Lower measurement accuracy
- ✗ Less sensitive for small cell imbalance detection

ADS1115 External ADC



Pros:

- ✓ 16-bit ADC resolution
- ✓ More accurate voltage readings
- ✓ Better for detecting small cell differences
- ✓ More stable measurement performance

Cons:

- ✗ Requires extra hardware
- ✗ Extra wiring through I2C
- ✗ Slightly more software setup
- ✗ Does not replace the Arduino Mega

Category	Arduino Mega (Built-in ADC)	ADS1115 External ADC
 Resolution	10-bit	16-bit
 Connection	direct analog pins	I2C to Mega
 Role	controller + basic ADC	external measurement upgrade
 Best use	prototype / basic logging	accuracy upgrade

 ADS1115 substitutes the Arduino Mega's analog voltage reading for those channels; it does not replace the Arduino Mega itself.

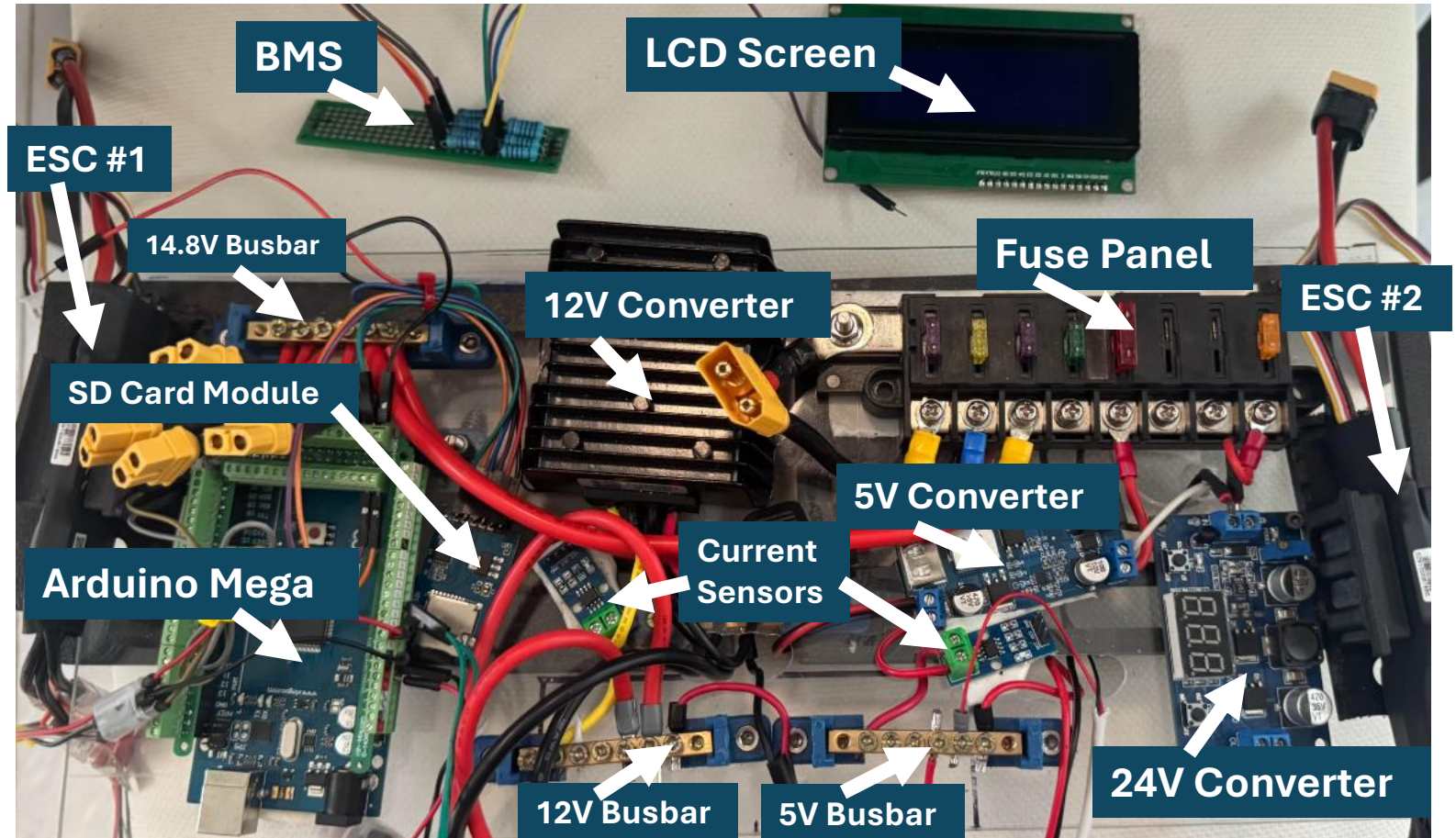
Power Distribution Board

Current Layout Challenges

- Component density makes troubleshooting difficult
- Long wire runs increase clutter and noise risk
- Power and signal wiring are mixed together
- SD card and telemetry components are harder to access

Planned Improvements

- Group components into power, control, and interface zones
- Shorten and label wiring with cable tags
- Use cable raceways, zip-tie mounts, and bread ties
- Add panel-mounted LCD



Power Distribution Board Layout Redesign (Draft)

Voltage Divider Board



12V Converter



12V Busbar



24V Converter



5V Converter



5V Busbar



Fuse Panel



ESC #1



ESC #2



Current Sensor



Temp. & Humidity Sensor



Current Sensor



Data Logger

Arduino Mega for Telemetry



- Battery Testing,
Performance & Wheels



Battery Testing, Performance & Wheels

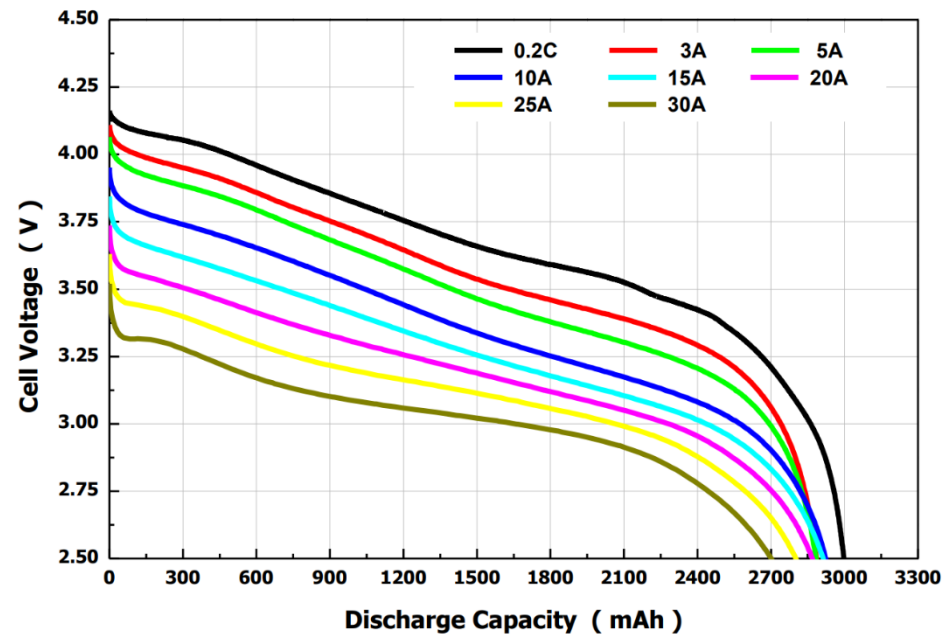
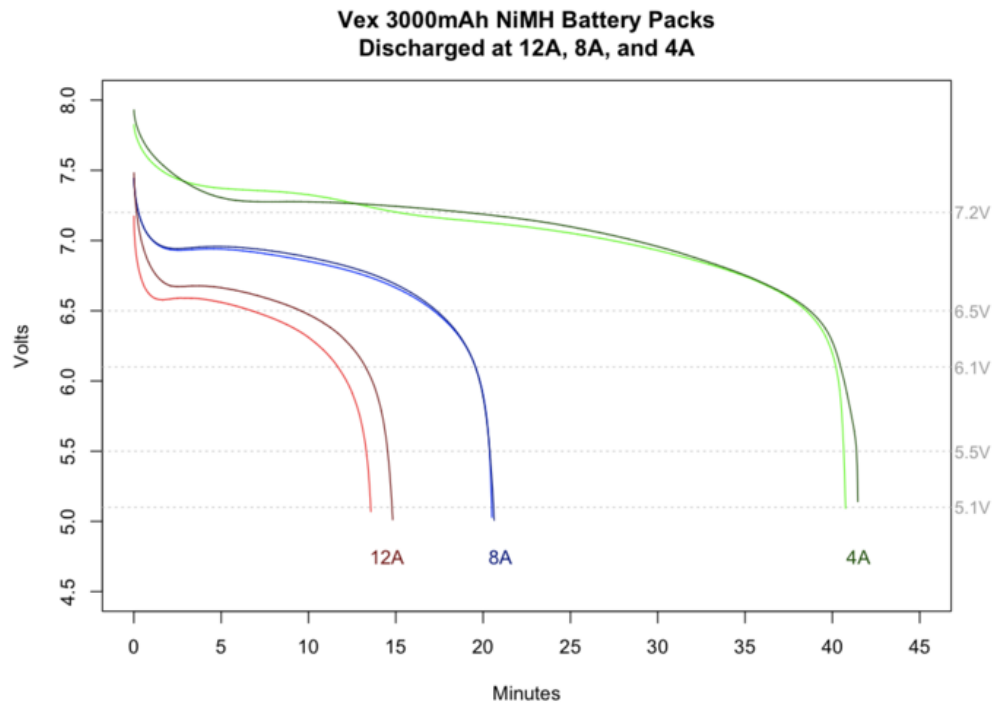
Problems:

- The high consumption on the system leads to the battery to run out faster than expected. (14.8V -> 12V)
- The battery's performance has not been tested or properly documented. (Voltage vs Time, Charge vs Time & Current) Curves
- The wheel motors consume more current than estimated which blow their fuse. And do not have solid cable connections.
- The digital connections between the software suite and ESCs for the wheels were hard to work with.



Battery Curves

To measure the performance of a battery used in the rover, constant loads can be connected to it for an extended period of time. At different current values the time it takes to go from its nominal value to 12V.



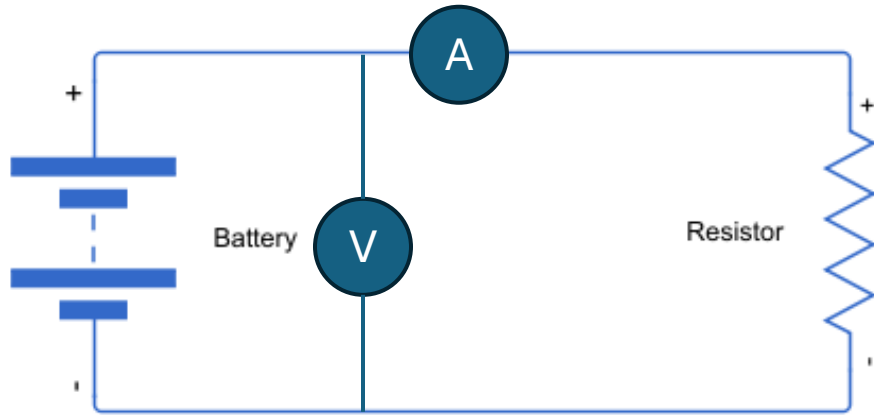
Battery Curves

- **The measurement can be done via the sensors, and the load can be represented through either a motor or the discharge mode in a battery charger.**
- **Dynamic curves that try to emulate a real-life ROVER run can be done with the use of a constant load in addition to a dynamic load like a motor.**
- **Initial curves can be simulated via softwares like MATLAB/Simulink or Ltspice to get theoretical values.**



Battery Curves

Static Load

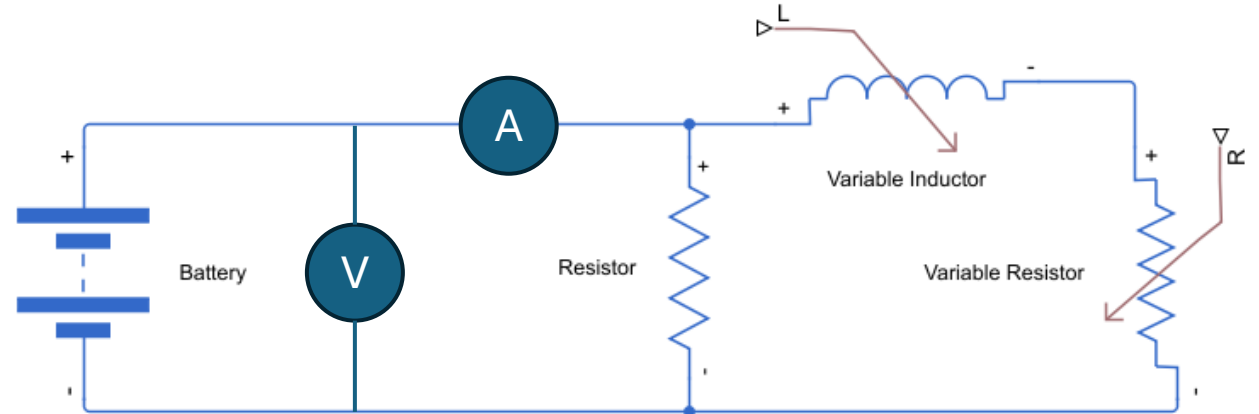


Ammeter

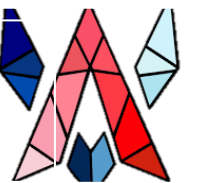


Voltmeter

Dynamic Load



***** These are not the proper simulation models, they are for electrical clarity*****



State of Charge (SoC) of Battery

- To know the exact state of charge according to the measured cell voltage, look for a manufacturer's specification sheet.
- If information is not readily available a good rule of thumb for High-Voltage (HV) and regular LiPo batteries is:

LiPo Cell Voltage (Approximated*)	State of Charge (SoC %) per cell	Notes
4.20 +	100 %	Fully charged. Not safe for storage.
4.10	90 %	-
4.0	80 %	Ideal for non-long-term storage (daily)
3.9	60 %	
3.7 – 3.85	40 – 50 %	Ideal for long-term storage
3.5	20 %	
Below 3.3	Approaching 0 %	Not safe to completely discharge cell



Solutions for Battery Supply

- These solutions apply in the case the battery is the only component that can be changed. Or more preferably in conjunction with other mitigations in the rest of the traction system.



Battery Performance

To resolve the issue of the battery not being able to meet the demands of the system, a couple of solution may be applied to the battery without changing other components in the ROVER.

1. Acquiring a battery with a larger charge capacity and same voltage (LiPo 22000mAh, 14.8V, 4S)

- **Pros:** Not a significant additional space requirement. Takes longer to discharge.
- **Cons:** Not easily available or cheap.

2. Putting 2 batteries in parallel with a protective system for back feeding (LiPo 44000mAh, 14.8V, 4S)

- **Pros:** Would meet all possible energy demands of the ROVER. Second battery could provide backup in case of a disconnection.
- **Cons:** Not space efficient. Purchase of a second battery with the same characteristics.



Battery Performance

To resolve the issue of the battery not being able to meet the demands of the system, a couple of solution may be applied to the battery without changing other components in the ROVER.

3. Acquiring a battery with a larger charge capacity and larger voltage (LiPo 30000mAh, 22.2V, 6S)

- **Pros:** Takes longer to discharge. ***12V Converters would not turn off during operation when charge is low**.*
- **Cons:** Not easily available or cheap. Significant additional space requirement. Would need an additional DC converter to meet the voltage of the wheel motors.



Solutions for Battery Demand

- These solutions apply in the case the loads are the only component that can be changed. Or more preferably in conjunction with other mitigations in the rest of the traction system.



Load Performance

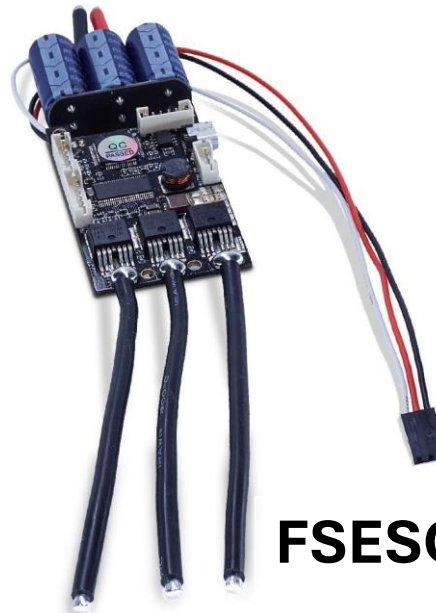
To resolve the issue of the battery not being able to meet the demands of the system, a couple of solution may be applied to the loads in the wheel system without changing other components in the ROVER.

1. Acquire a Vedder Electronic Speed Controller (VESC) to replace ESCs and limit the current (MakerX, Flipsky)

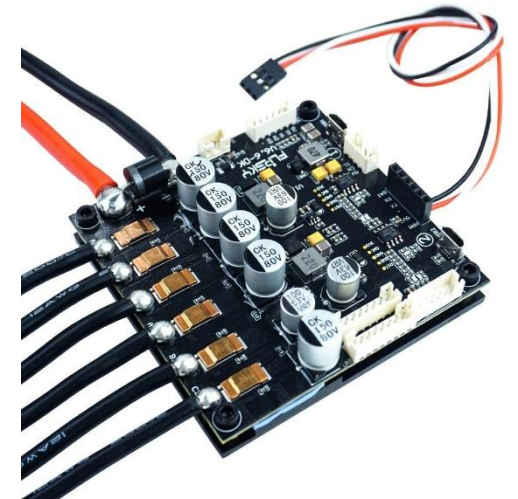
- **Pros:** Current can be limited to a specific value. Motor control can be centralized to a single component. C,C++/Python for external control. VESC Tool for interface & configuration.
- **Cons:** Not cheap. In most cases 1 module per 2 motors. Additional coding is required. [MakerX = 165\$] [Single Flipsky = 75 \$] [Dual Flipsky = 130\$ - 230\$]



Go FOC DV4s



FSESC4.12

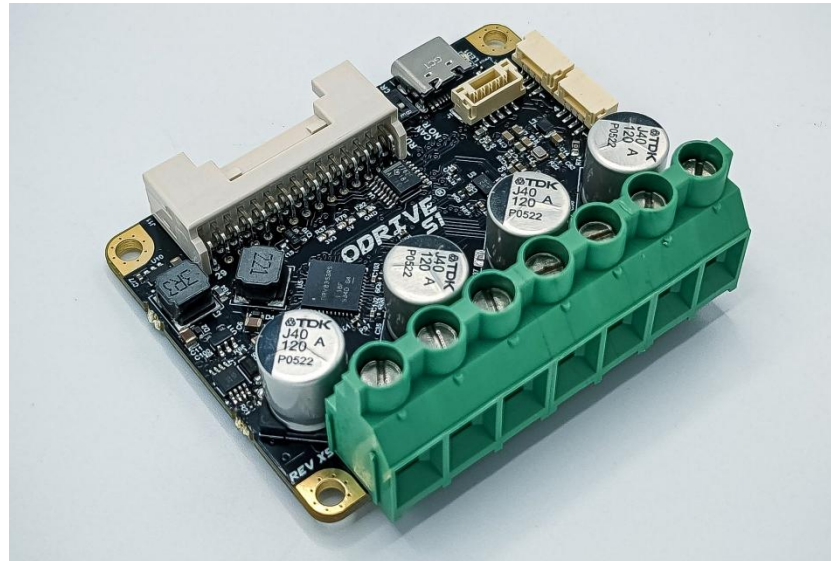


FSESC6.6
FSESC4.2

Load Performance

To resolve the issue of the battery not being able to meet the demands of the system, a couple of solution may be applied to the loads in the wheel system without changing other components in the ROVER.

1. Acquire an ODrive S1 to replace ESCs to limit current and supply constant torque (ODrive)
 - **Pros:** Can have many limiting modes. Small and space efficient. C,C++/Python for external control. ODrive GUI for interfacing.
 - **Cons:** In most cases 1 module per motor. Not so cheap. Additional coding required. [150\$]



Load Performance

To resolve the issue of the battery not being able to meet the demands of the system, a couple of solution may be applied to the load in the wheel system without changing other components in the ROVER.

3. Keep the same system just limit the mechanical load that increments the current

- **Pros:** No significant change to the electrical system. No additional component than can fail and disconnect the system.
- **Cons:** Susceptible to current swings and no precise control of the wheels.



Wheel Connections

Power connections



MR30 & MR40 Connectors

- High ampacity = 30A continuous
- Not so simple to disconnect
- Soldering is required



Wheel Connections

Power connections



Bullet Brass Connectors

- High ampacity = 20A continuous (2mm)
- Simple to disconnect
- Soldering is required
- Compatible with current ESC & Motors



Wheel Connections

Power connections



Push-In & Click Connectors

- High ampacity = 20A continuous
- Simple to disconnect
- **Can be other models as well**
- Scolmore CT205C

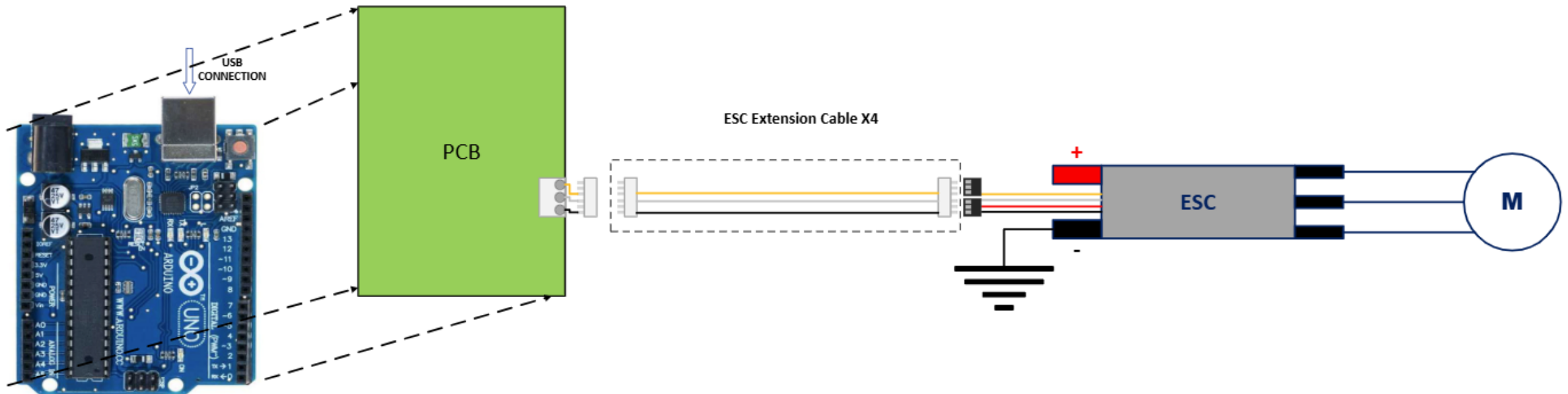


Wheel Connections

Use PCB module for Arduino

Digital connections

- Design a small PCB extension for the Arduino that controls the digital communication of the ESCs.
- GND, **Signal** & **Direction** cables are joined by a single 3 cable connector at the module and at the ESC.



-
- +
 -
 -

Robotic Arm Suite



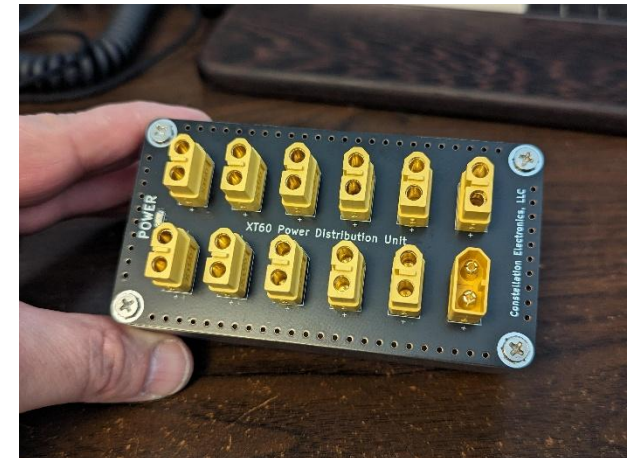
Robotic Arm Suite

Optimizations:

1. Connector panel
2. Change motor drivers
3. Switch converter for a voltage regulator
4. PCB integration

1. Connector panel

- Make a hole in the ceiling and screw this mount in place.
- Route the motor outputs from the robotic arm suite through the hole and connect them to the mount.
- Then, from the mount, connect extension cables to the motors.



Anderson powerpole pp15-45

- New connector: Anderson powerpole pp15-45
- Max current rating 45A per connector
- Easier to disconnect in comparison with XT90, which can be a downside but when lifting the rover's top panel, it's an advantage



XT90

- Max current rating 90A per connector
- Hard to disconnect when lifting the rover's top deck
- Requires soldering and can break when disconnecting



Connectors

Connector	Anderson Powerpole	XT90
Width	7.9mm	20.8mm
Height	7.9mm	10.5mm
Length	40mm	47mm
Cable selection	10-12 AWG	8-10 AWG
Max. current	45 A	90 A
Assembly method	Crimped	Soldered

Molex micro fit

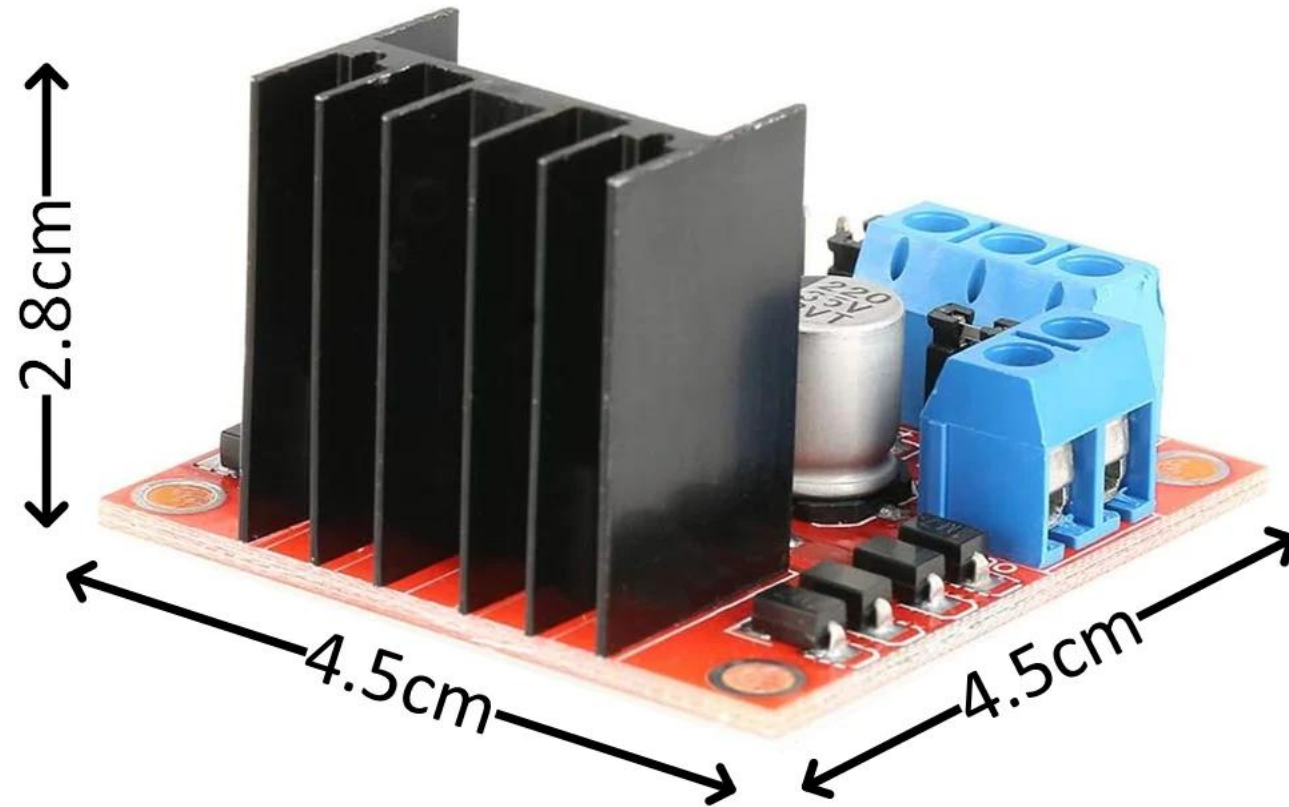
- Good replacement for micro servo motor connector
- 5 A per pin (servo stall current = 650mA)
- Strong latch (click lock)
- Designed for vibration environments
- Assembly method: crimped



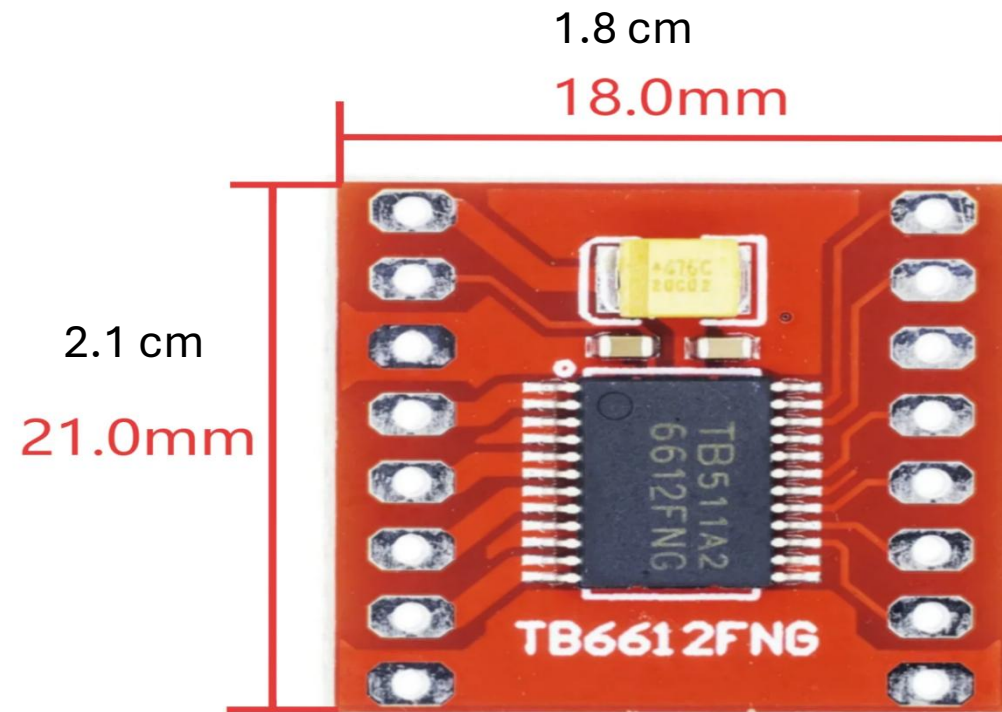
2. Motor Drivers

Driver	Voltaje motor (V)	Corriente continua (A)	Corriente pico (A)	Voltaje lógico	Tipo transistor	Caída de voltaje	Canales	Heat sink	Precio (USD)
L298N (current)	5 – 35	2	3 A	3.5 – 5	BJT	Aprox. 1.8 V	Dual	Sí	3 – 6
DBH-12 (Also recommended)	5 – 14.5	20	30	3.3 – 12	MOSFET	Aprox. 0.3 V	Dual	Sí	15 – 25
TB6612FNG (recommended)	4.5 – 13.5	1.2	3.2	2.7 – 5.5	MOSFET	Aprox. 0.5 V	Dual	No	3 – 5
DRV8871	4.5 – 13.5	1.2	3.2	2.7 – 5.5	MOSFET	Aprox. 0.6 V	Single	No	4 – 7
MX1508	2 – 10	1.5	2.5	1.8 – 7	MOSFET	Aprox. 0.5 V	Dual	No	1 – 3
BTS7960	6 – 27	20	43	3.3 – 5	MOSFET	Aprox. 0.2 V	Single	Sí	7 – 15
VNH5019 (Also recommended)	5.5-24	12	30	3.3 - 5	MOSFET	Aprox. 0.1-0.3V	Dual	Only with high current	30-60

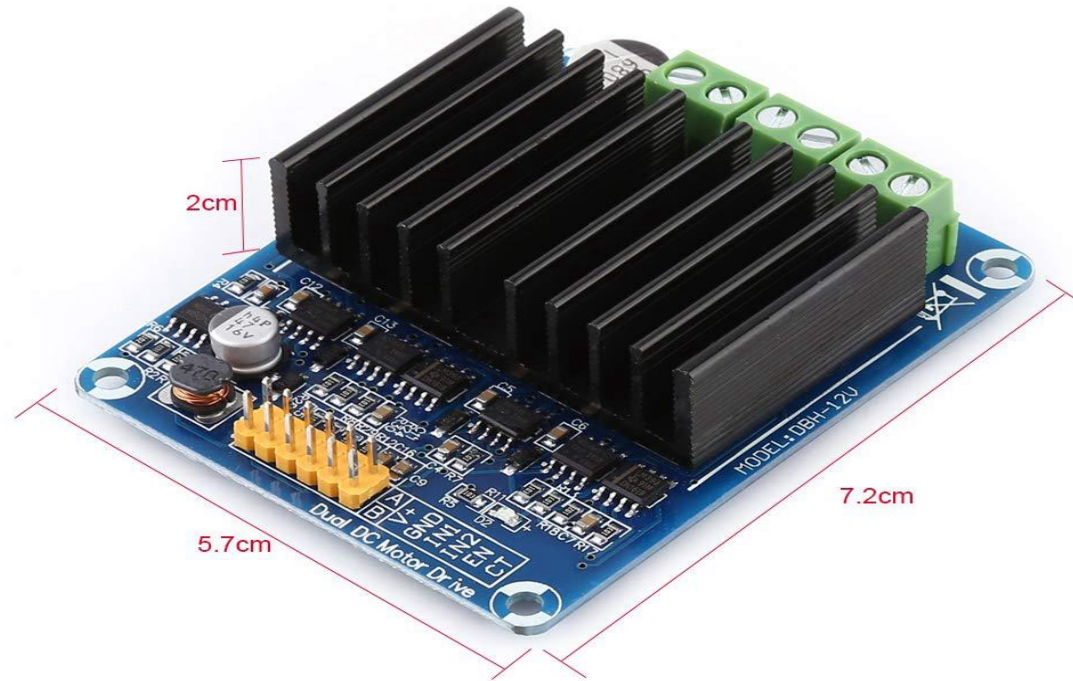
IC L298N



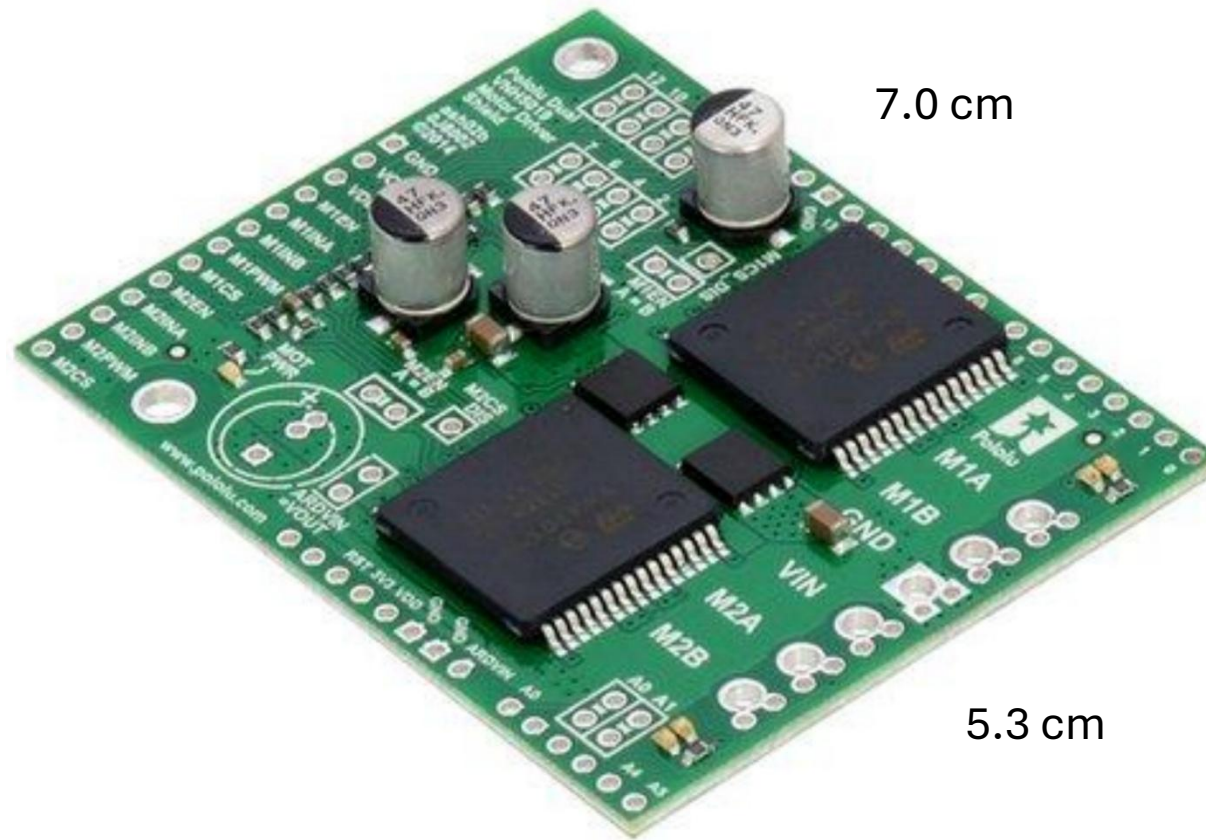
IC TB6612FNG



IC DBH12



IC VNH5019



7.0 cm

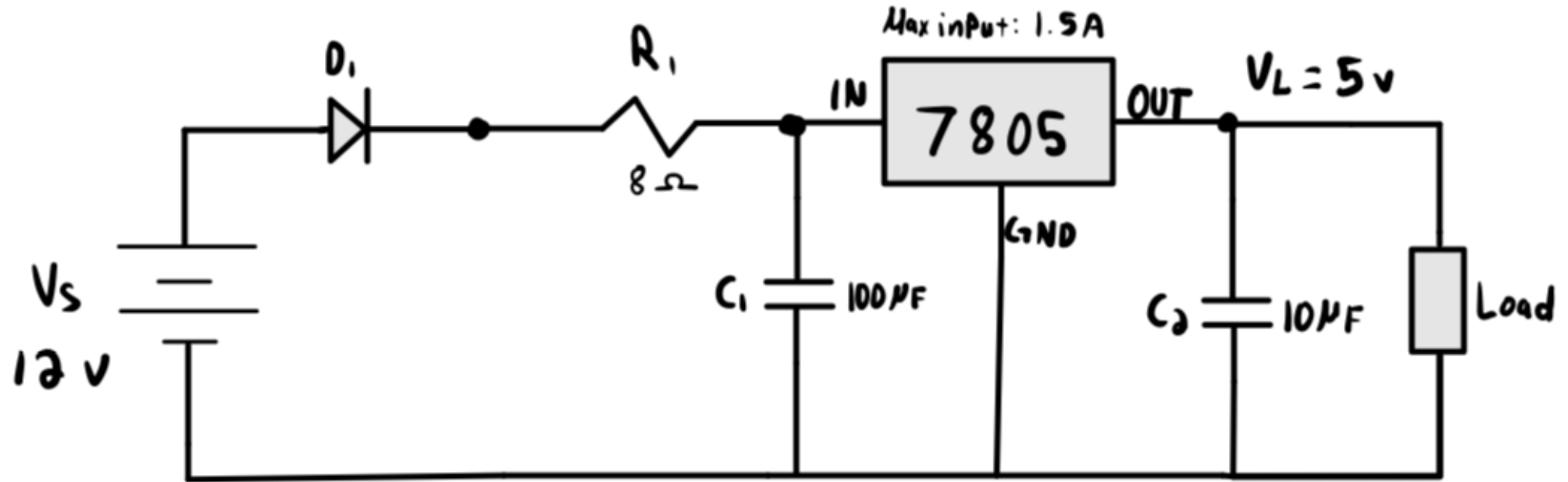
5.3 cm

3. Change DC converter to Voltage regulator

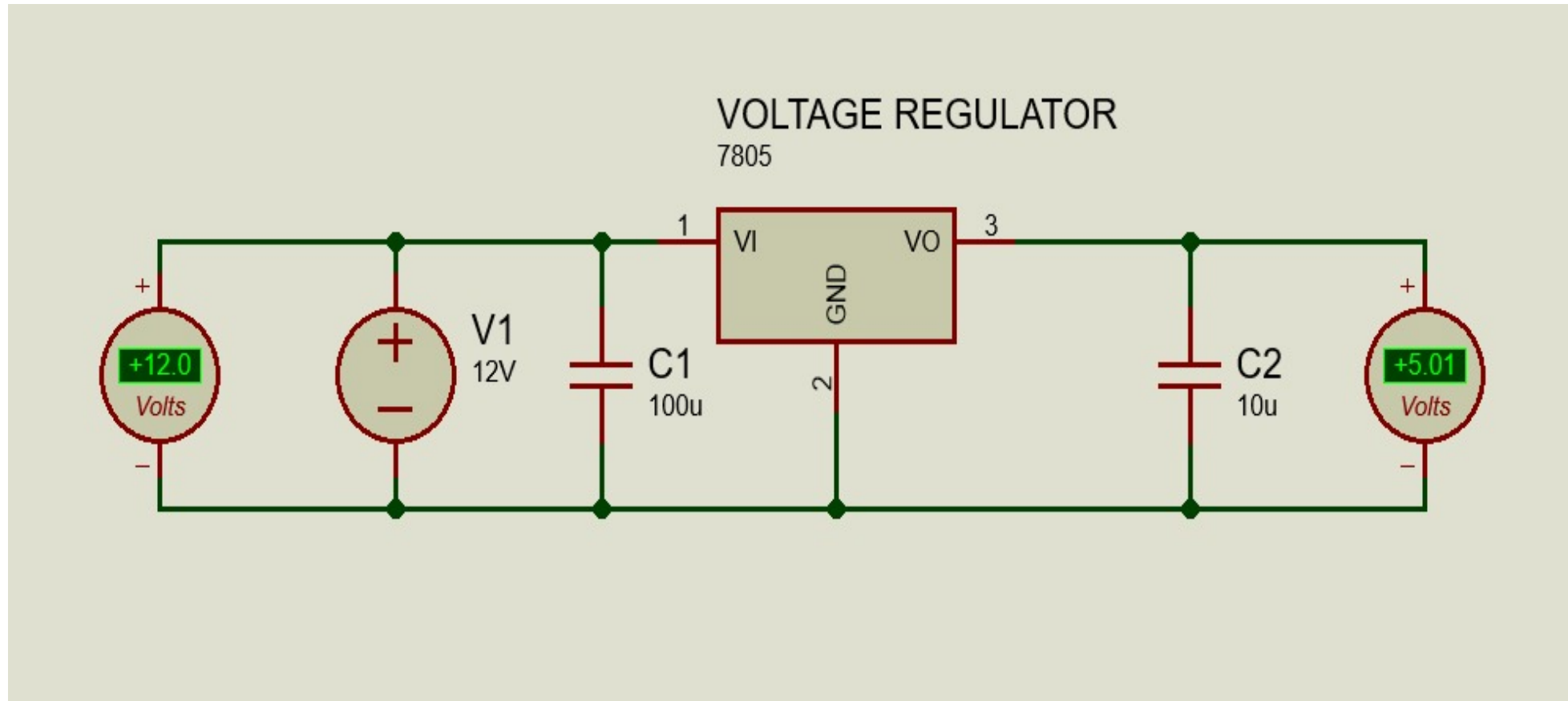
- Max current rating: 1.5A
- Servo stall current: 650mA
- Smaller than the converter while doing the same function
- Handles 7 – 35V as input voltage
- Can supply the motor driver's 5 V logic input, as each driver only draws about 5–15 μA



Schematic

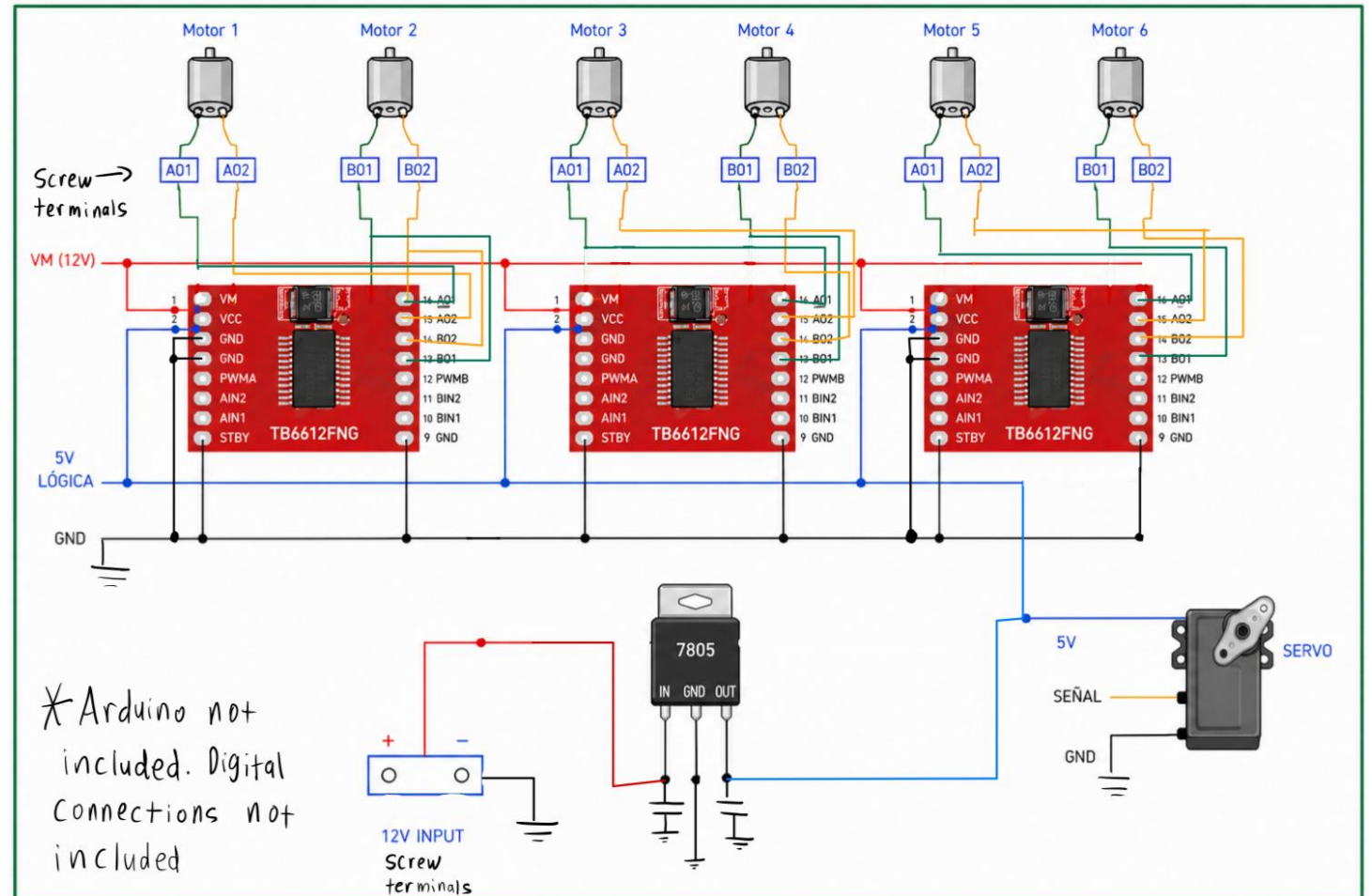


IC 7805 Simulation



4. PCB Integration

- Integrate Arduino Mega, 7805 (with input and output capacitor), and 3 TB6612FNG to a single PCB.
- Use screw terminals to connect the cables for motor output.
- Use screw terminals for the 12V input and GND.



- +
-
-

Wiring Layout & Design



Wire Sleeves



- Color Coding

- Stripes of colored tape will be used to identify cables.
- Color will now be assigned by system and not by division.
- Example:
 - Wire that goes from system A (Red) to system B (Blue) will be striped with red and blue.
 - Wire with more than two destinations will carry each destinations color, (RED + Blue + Yellow).

- Accessibility

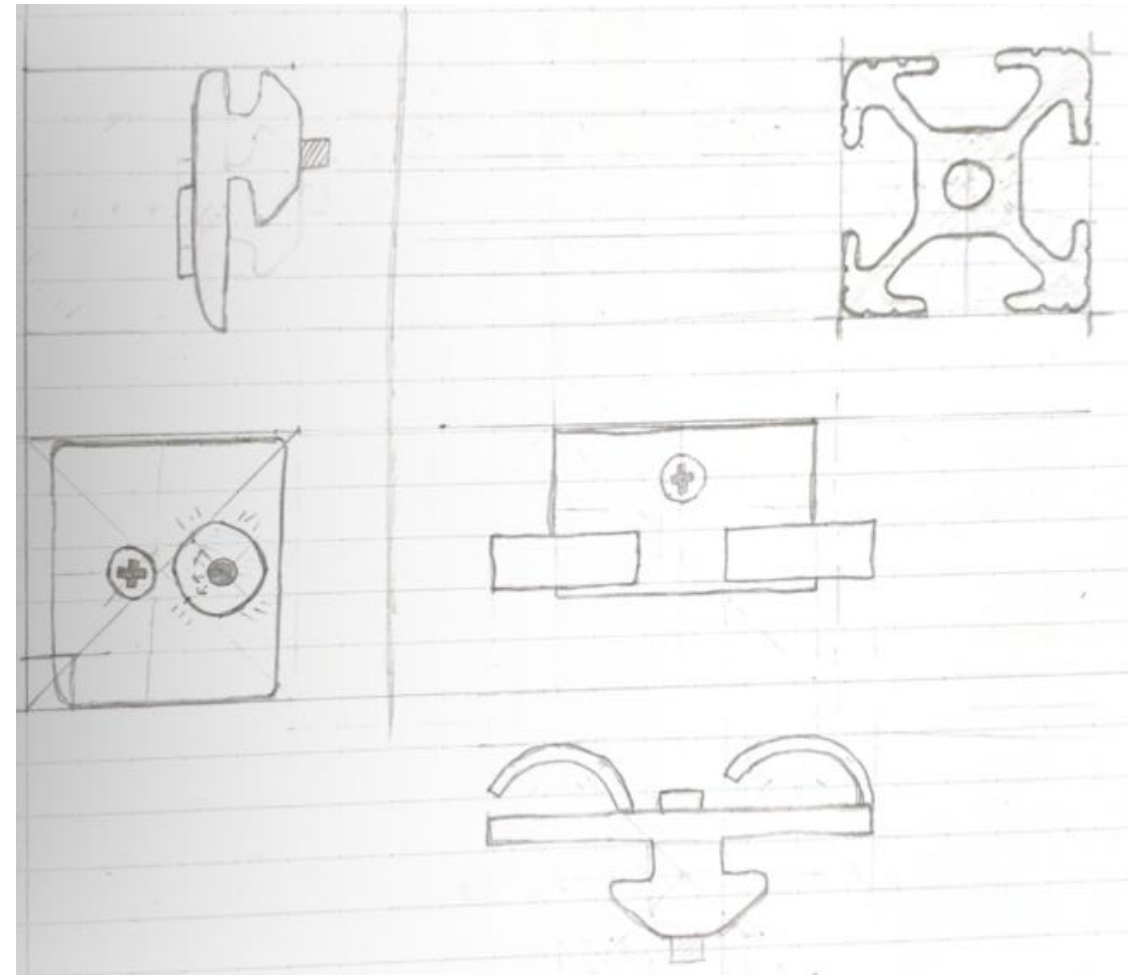
- Less of a size limit with cable diameter.
- Accessible for maintenance after installation.





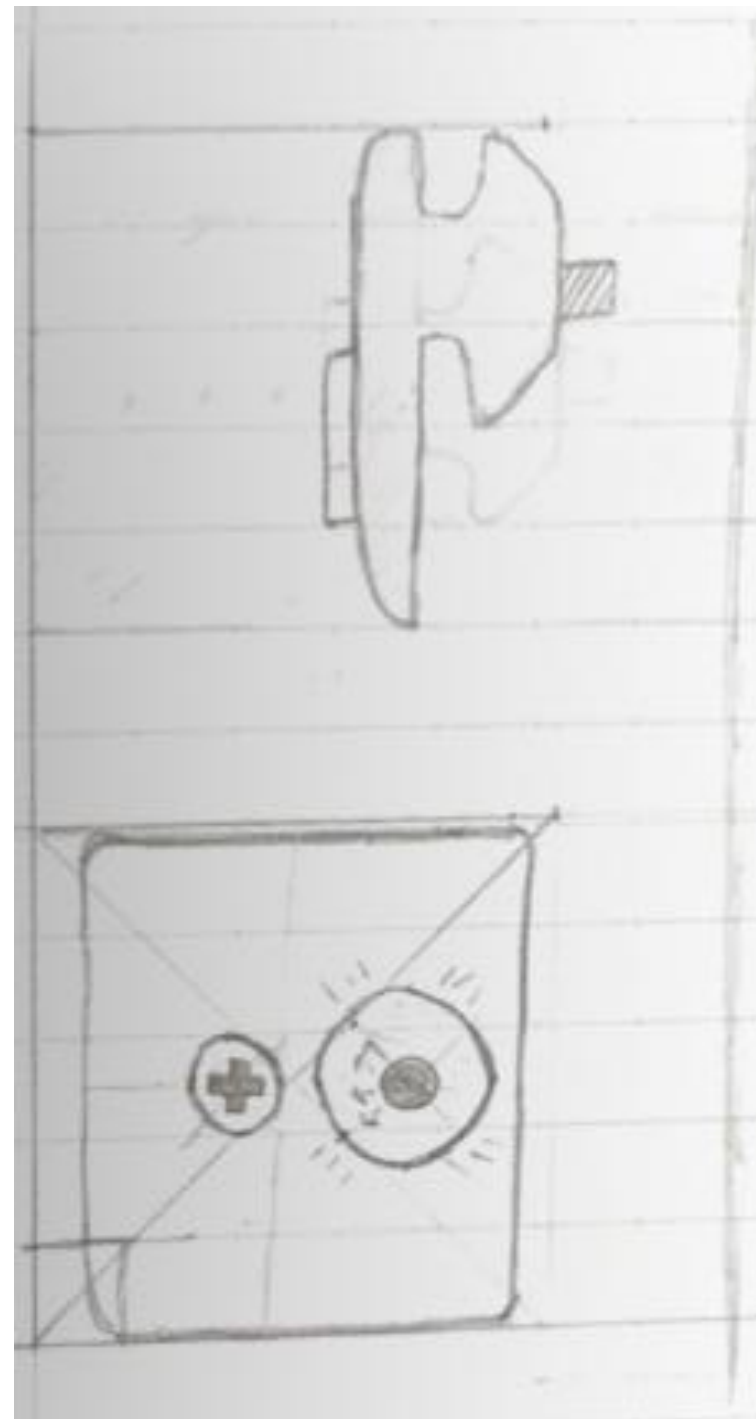
Holds

- Holds must be rail mounted, with the wire being accessible.
- Due to limited wire space to move along the rover, holds must be able to hold at least 2 separate sleeved wires.
- Holds could be made with 3d printed hooks or Velcro along the 3d printed frame.



Lighting for maintenance

- Rail mounted LED flashlights, wired in series with one or two for each board tier.
- Manually turned on or off with switch



Wire tension

To keep a sleek look and design:

- Wires must be kept in slight tension to stay along the desired direction and not interfere with tinkering.



+
•
○

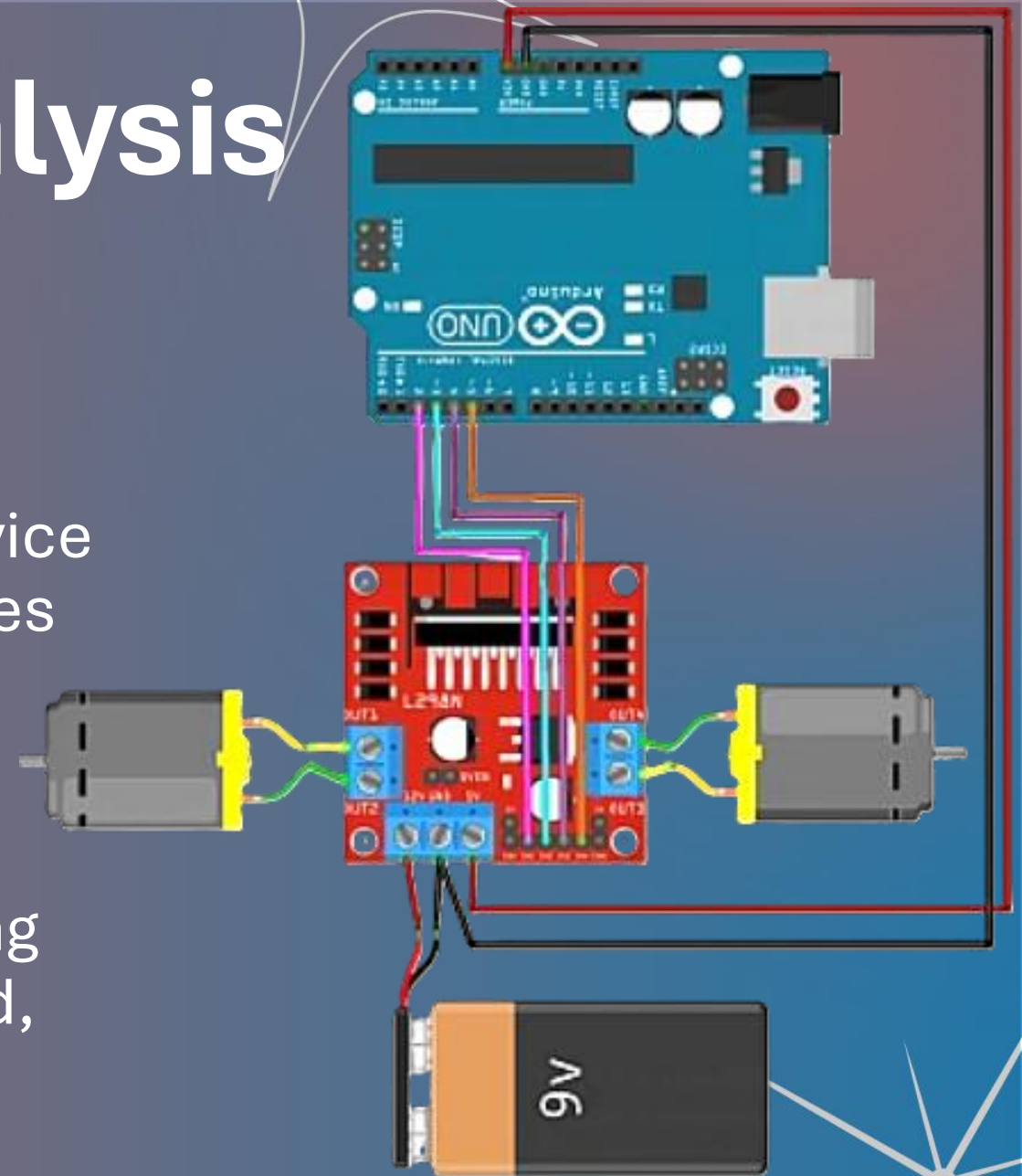
Science Suite



1. Motor Driver Analysis

What is a Motor Driver?

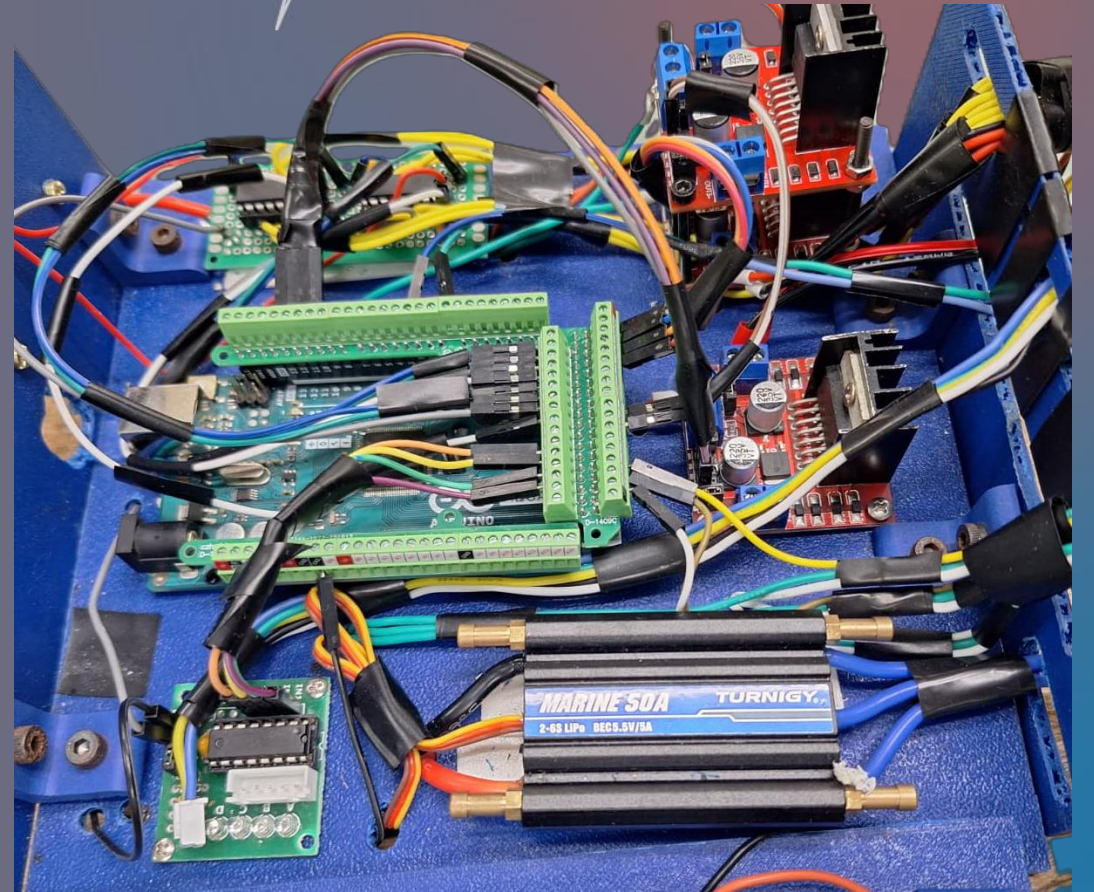
- A **motor driver**, also known as a control motor, is an electronic device or module that controls and manages the operation of an electric motor.
- It serves as an interface between a microcontroller or other control system and the motor itself, enabling precise control of the motor's speed, direction, and other parameters.



The Main Problem

Space and Power Efficiency

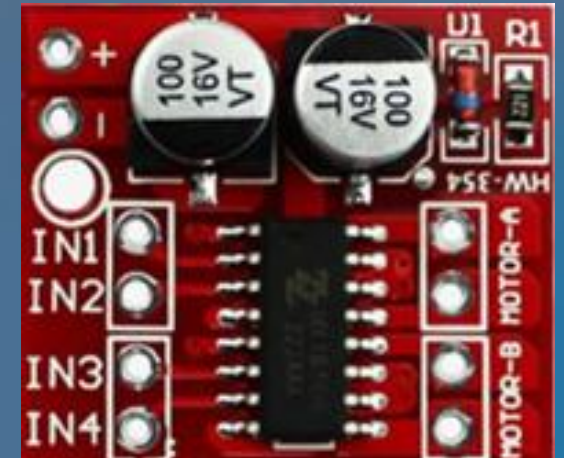
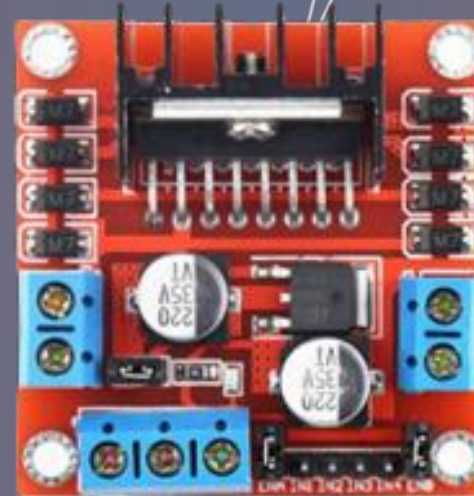
- In the current setup, the Science Suite relies on bulky and inefficient motor drivers, that can pull higher currents than needed, dissipate great amounts of unused power, and take up large amounts of space.
- While current doesn't present a grand issue right now, space efficiency does.



Solution

Compare, contrast, and choose

- The idea is to compare the 4 most common motor drivers in both the modern and legacy market, see their differences and make an informed decision on which motor driver is best suited for the Science Suite.



Comparison: Shared Parameters

Absolute Maximums

Parameter	L298N	L293D	TB6612FNG	MX1508
Motor supply rating (V)	50	36	15	10
Logic supply rating (V)	7	36	6	7
Input voltage rating (V)	-0.3 to 7	7	-0.2 to 6	7 max, referenced to VCC
Output voltage rating (V)	50	-3 to VCC2 + 3	15	10 max, referenced to VDD
Continuous output current per channel (A)	2	0.6	1.2	Channel A = 0.8 Channel B = 1.5
Peak output current, short pulse or nonrepetitive (A)	3 at 100ms	1.2 at 100µs	3.2 for a 10 ms single pulse	1.5
Peak output current, repetitive or continuous pulse (A)	2.5 repetitive	N/A	2.0 for a 20ms continuous pulse (duty cycle ≤ 20%)	2
Power dissipation (W)	25	N/A	0.78 IC only 0.89 and 1.36 with PCB copper area	1.5
Operating temperature (°C)	-25 to 130	0 to 70	-20 to 85	-20 to 85
Storage temperature (°C)	-40 to 150	-65 to 150	-55 to 150	-55 to 150
Thermal shutdown temperature (°C)	N/A	N/A	175	150
Electrostatic Discharge HBM (V)	N/A	±2000	N/A	3000
Thermal resistance θ_{JA} (°C/W)	13 (PowerSO-20) 35 (Multiwatt15)	36.4	160 (IC only)	80

Comparison: Shared Parameters

Recommended Operating Ranges

Parameter	L298N	L293D	TB6612FNG	MX1508
Motor supply recommended usage (V)	up to 46	4.5 to 36	2.5 to 13.5	2 to 9.6
Logic supply recommended usage (V)	4.5 to 7	4.5 to 7	2.7 to 5.5	1.8 to 5
Input high voltage (V)	2.3 to VSS	2.3 to VCC1 (or 7 when VCC1 ≥ 7)	≥0.7×VCC, up to VCC+0.2	2 min
Input low voltage (V)	-0.3 to 1.5	-0.3 to 1.5	-0.2 to 0.3×VCC	0.8 max

Electrical Characteristics

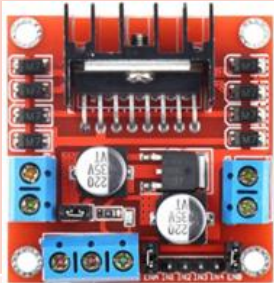
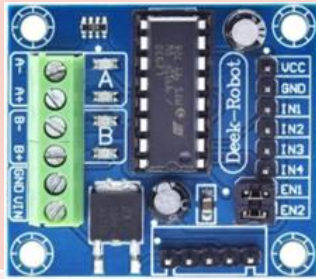
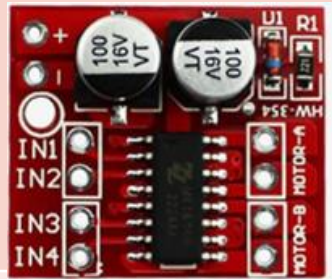
Parameter	L298N	L293D	TB6612FNG	MX1508
Output voltage drop / saturation / on-resistance	VCEsat total drop: 1.80 typ / 3.2 max at 1 A 4.9 max at 2 A	VOH: VCC2-1.8 to VCC2-1.4 VOL: 1.2 to 1.8	0.5 typ / 0.7 max at 1 A; 0.15 typ / 0.21 max at 0.3 A	RON1 = 0.49 / 0.53 Ω; RON2 = 0.49 / 0.53 Ω
Internal pull-down resistance (kΩ)	N/A	N/A	200	12
Thermal shutdown hysteresis (°C)	N/A	N/A	20	20
Rise time tr (ns)	N/A	300 / 100	24	N/A
Fall time tf (ns)	N/A	300 / 350	41	N/A

Comparison: Shared Parameters

Driver Sizes

Parameter	L298N	L293D	TB6612FNG	MX1508
Module Measured	L298N dual full-bridge motor driver module	Deek-Robot Mini 2-Channel Motor Drive Module with L293D	TB6612FNG dual motor driver module / breakout	MX1508 dual H-bridge motor driver module
Approximate Module Size (mm)	43 x 43 x 27	39 x 35	21 x 18.5	24.7 x 21
Approximate Chip Size (mm)	19.6 x 4.25 x 24.35	19.8 x 6.35	7.6 x 7.8 x 1.2	9.9 x 6 x 1.4
Pin count	15	16	24	16

Images and Datasheets

Parameter	L298N	L293D	TB6612FNG	MX1508
Image				
Data Sheet	www.st.com	www.ti.com	toshiba.semicon-storage.com	cdn.hackaday.io
Full Excel Sheet:	MotorDriverComparison			

Comparison: Shared Parameters

Analysis

Parameter	L298N	L293D	TB6612FNG	MX1508
Pros	• Handles higher voltage than the other three here. • Supports two channels and current sensing. • Useful for motors and steppers. • High-voltage capability in this group. • Current-sense access is built in. • Strong fit for DC motors and steppers.	• Easy to use and widely understood. • Built-in protection for inductive loads. • Good for simple low-power motor control. • Built-in output clamp diodes reduce external parts. • TTL-compatible inputs and inductive-load support. • Broad, familiar legacy support.	• Efficient, compact, and practical for modern small motor systems. • Good PWM support and built-in protection. • Low heat compared with Darlington drivers. • MOSFET output stage gives low output loss. • Low standby current and low quiescent current. • Built-in dead time and low-voltage detection.	• Lightweight and battery-friendly. • Low power draw and simple dual-H-bridge use. • Well suited to compact hobby systems. • Very low standby and quiescent current. • Built-in anti-common-conduction protection. • Thermal protection and 3 kV HBM ESD rating.
Cons	• Inefficient compared with MOSFET drivers. • Can run hot under load. • Needs more support around the power stage. • Saturation loss is large, so efficiency is low. • External clamp diodes are needed for inductive loads in typical use. • Quiescent current is relatively high.	• Lower current headroom. • Larger voltage drop than MOSFET-based devices. • Not ideal when efficiency matters. • Current capability is modest compared with the stronger parts here. • Darlington output stages waste voltage under load. • Less suitable for higher-current motors.	• Not a high-voltage part. • Thermal design depends heavily on the PCB. • Not aimed at large motors. • Lower voltage ceiling than L298N and L293D. • Better for small motors than heavy loads. • PCB copper area matters a lot for heat.	• Not a high-voltage solution (<10V). • Best treated as a small-motor driver, not a broad industrial part. • Less documentation depth than the other three (everything is in chinese). • Narrower operating range than the larger drivers. • Current ratings are split by channel/use case, which makes comparison less uniform. • Less complete characterization than the TI and ST parts.

2. Scoop Motor analysis

What can it do?

- The scoop motors currently use a HSR-1425CR, a continuous-rotation servo, not a true position servo.

What should it do?

- The scoop motors must **reliably** go from 0° , rotate 270° clockwise, and return back to 0° .
- It must have repeatable angle control, which means it must retain positional feedback at all times, something a speed-only actuator cannot do.



The Main Problem

Direction vs Position

- Due to the servo being directional, it regularly misaligns, overshoots, undershoots, and is in general, wholly unreliable.
- Any change in load immediately changes its speed (recall Newton's Second Law).

Positional Feedback

- Even if continuous-directional servos allowed for controlled position, any natural misalignment or drift can skew the neutral point of reference, which is why positional feedback is needed in order to correct any drift or misalignments.

Possible Solution: Closed-Loop Stepper

Why?

- A closed-loop stepper, alongside an encoder and a homing switch, can give commanded positions and use encoder feedback to avoid missed steps.

Possible Considerations

- With a variety of closed loop steppers online, more research is needed to determine the best suited one; however, the [Nema-17](#) seemed a popular choice.

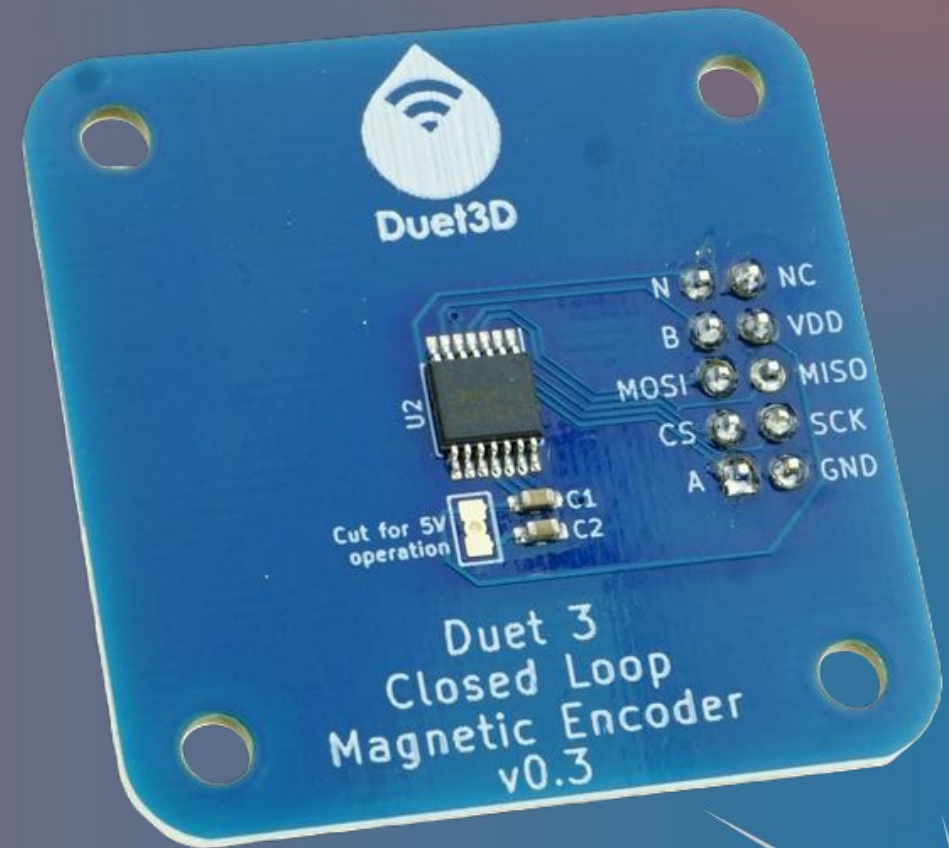
Note: specific details pertaining to electrical compatibility will have to be further researched



Possible Solution: Closed-Loop Stepper

What is an Encoder?

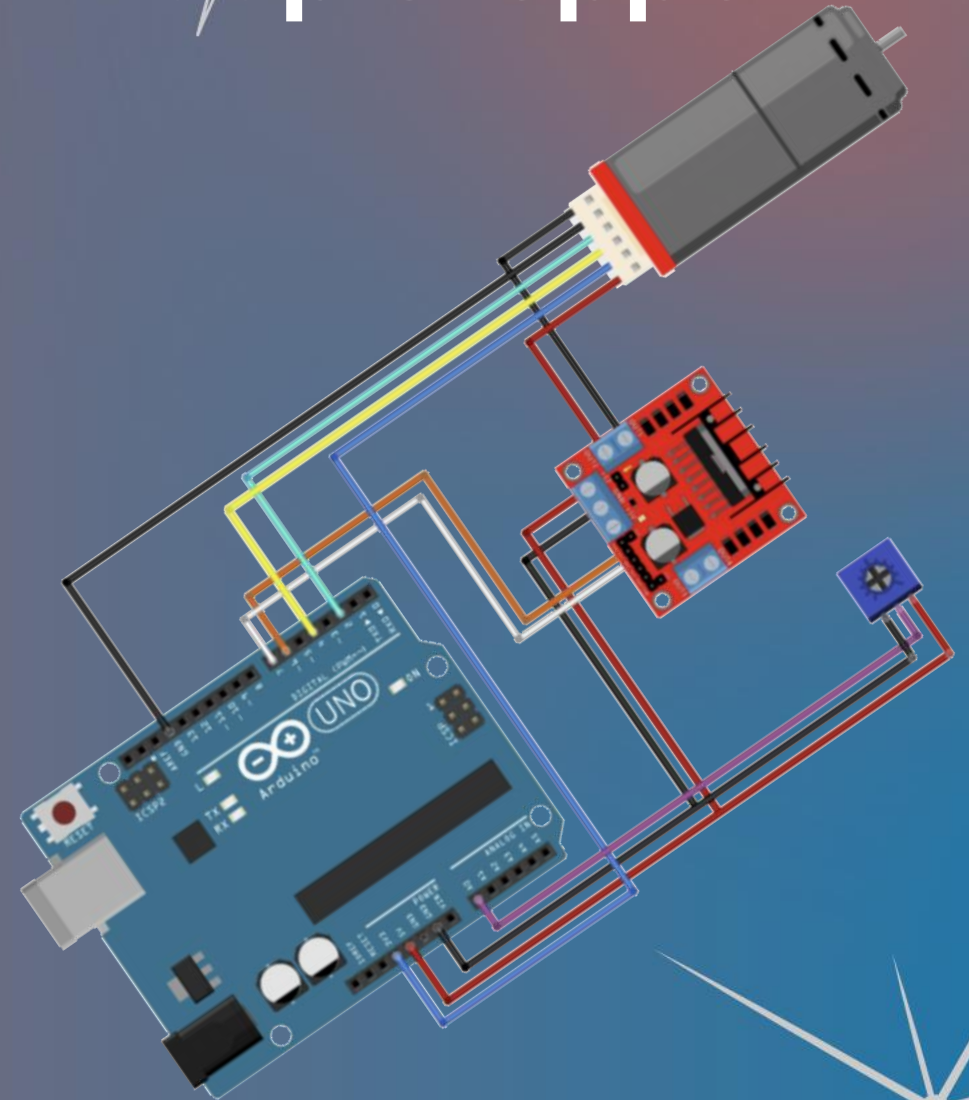
- The main function of the stepper motor encoder is to measure the rotation angle and speed of the motor.
- When the motor is moving, the encoder will detect the position of the motor rotor through the sensor and convert it into a digital signal.
- These digital signals can be used to control the movement of the motor, thereby achieving position control and speed control.



Possible Solution: Closed-Loop Stepper

What is a Homing Switch?

- A homing switch is placed in a predetermined position to provide a reference point. Once it is activated the motor “knows” where it is.
- From there you can use the precision of the stepper motor to accurately move into any position you wish.
- A **Hall Effect switch** can respond to magnetic fields, so they can be triggered by a small magnet (i.e., an encoder).



Possible Solution: Smart Servo

Why

- Smart servos and smart actuators, while sometimes expensive, can be compact and work best for “move-and-hold” motions.
- Smart servos combine a motor, gearbox, encoder, and controller into one unit, offering sophisticated feedback (position, speed, temperature) and programmable functionality.
- They give feedback and are used in various high-end applications.

Possible Considerations

- <https://www.seeedstudio.com/Feetech-ST-3215-C044-Heavy-Duty-Servo-7-4V-1-191-Gear-Reduction-p-6460.html>
- <https://www.hiwonder.com/products/lx-16a>
- <https://www.robotis.us/dynamixel-xm430-w210-t/> <-Expensive!

